Michelson Interferometer for Correction of Translation Stage Deviation

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Michelson-Interferometer zur Korrektur von Abweichungen eines Translationstisches

Bachelorarbeit

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vorgelegt von

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ABSTRACT

This thesis presents the development of an interferometric measurement system for the calibration of an automated translation stage used in a field-resolved infrared spectroscopy (FRIS) experiment based on electro-optic sampling (EOS). The work was motivated by two limitations of the currently used control software: a limited positioning accuracy, and the inability to detect small residual forward and backward movements of the stage when it is commanded to remain at a fixed position. The main goal of this thesis was to address the first limitation; to this end, a Michelson interferometer setup and corresponding analysis software were developed to improve the measurement of the stage displacement.

First, different laser sources were characterized and compared with respect to wavelength, beam quality, and intensity stability. Based on these measurements, the Thorlabs CPS780S laser was selected as the light source for the interferometer.

Two methods for detecting the interference signal were compared: optical detection with a camera and electronic detection with a photodiode. Software was developed to evaluate both signals and count the interference fringes. The comparison showed that the camera-based detection was more stable and more reliable for the investigated setup.

Since a conventional Michelson interferometer does not provide information about the direction of motion, a quadrature homodyne interferometer was additionally developed to address the second limitation and enable active correction of small stage movements around a locked position. This extension was successfully built and showed that both displacement and direction of motion can be determined from two phase-shifted detector signals. Although the resulting position-lock feedback loop was ultimately limited by the mechanical response of the translation stage and did not reach stable long-term operation, the direction of motion could be reliably determined, and the quadrature homodyne configuration was therefore selected as the detection method for the final integrated system.

CONTENTS

CHAPTER 1 INTRODUCTION

Advances in modern science are closely linked to the ability to perform precise measurements. Many scientific breakthroughs have only become possible because advances in experimental techniques enabled researchers to implement and verify new concepts with increasing accuracy. Therefore precise positioning systems are essential in many experimental and optical applications, where even small positioning errors can significantly influence measurement accuracy and system performance [1].

One of the most accurate and widely used methods for displacement and position measurements is optical interferometry [2]. It is based on the interference of coherent light waves, where changes in the optical path length produce measurable shifts in the interference pattern. Since the wavelength of light is on the order of hundreds of nanometers, interferometric techniques can resolve displacements that are only a small fraction of a wavelength, enabling sub-nanometer or even picometer resolution under suitable conditions [3].

One particularly demanding application is field-resolved infrared spectroscopy (FRIS) based on electro-optic sampling (EOS), which enables the direct measurement of molecular electric-field responses in the time domain. By recording the coherent response of vibrationally excited molecules, FRIS provides highly specific molecular fingerprints of biological samples such as blood plasma, opening new possibilities for biomedical diagnostics, health monitoring, and early disease detection, including cancer screening [4, 5].

In the experimental setup used for these measurements, a motorized translation stage introduces a controlled optical delay between a gate pulse and the measured infrared signal. Accurate knowledge of the stage position is crucial, as the reconstructed electric-field waveform and all subsequent spectral information directly depend on the temporal delay. The currently implemented control software, however, is affected by two distinct limitations. First, it only provides limited positioning accuracy and does not allow independent verification of the actual displacement. Second, when the stage is commanded to remain at a fixed position, small residual forward and backward movements occur that are not detected by the control software and would need to be actively compensated in order to keep the stage reliably locked in place.

The main goal of this thesis was to address the first limitation: an interferometer-based calibration method was developed to independently verify and improve the positioning accuracy of the translation stage. As an additional extension, a quadrature homodyne interferometer was developed to address the second limitation, since determining the direction of motion is a prerequisite for actively correcting such small drifts around a locked position.

CHAPTER 2

THEORETICAL BACKGROUND

Before discussing the experimental work, the theoretical foundations relevant to this thesis are introduced. The chapter begins with Gaussian beam optics and the beam quality factor M^2, since these concepts are important for characterizing the laser sources used in the interferometer. Afterwards, the basic principle of interferometry is explained using a Michelson interferometer, which forms the basis of the measurement setup. This is then extended to quadrature homodyne interferometry, because this method also allows the direction of motion to be determined. The following sections introduce the main components of the experiment: suitable laser sources, the operating principle and limitations of the electronic translation stage.

2.1 Gaussian Beam Optics

One objective of this thesis was the characterization of different laser sources. In many optical applications, laser beams are assumed to be Gaussian, with a beam profile that follows the Gaussian distribution. For this reason, Gaussian beam optics forms the basis for one important characterization parameter, the beam quality factor M^2[6].

2.1.1 Gaussian Beam

Laser beams are solutions of Maxwell's equations, and the Gaussian beam represents a particular solution. To understand the origin of this solution, we first consider the wave equation for monochromatic electromagnetic fields. In this case, the spatial field distribution is governed by the Helmholtz equation

$$\nabla^2 \tilde{E} + k^2 \tilde{E} = 0, \tag{2.1}$$

where $\tilde{E}$ is the complex field amplitude and $k = 2\pi/\lambda$ is the wave number [6].

In laser optics one is mainly interested in beams propagating predominantly along the z-direction with small transverse angles. Under this paraxial approximation the electric field can be separated into a rapidly oscillating carrier wave and a slowly varying complex envelope

$$\tilde{E}(x, y, z) = \tilde{u}(x, y, z)e^{-ikz} \tag{2.2}$$

where $\tilde{u}(x, y, z)$ is the slowly varying envelope, and e^{-ikz} represents the rapidly varying phase factor of propagation in the z-direction [7].

Substituting this expression into the Helmholtz equation and applying the paraxial approximation yields the paraxial Helmholtz equation

$$\nabla_T^2 \tilde{u} - i\,2k \frac{\partial \tilde{u}}{\partial z} = 0, \tag{2.3}$$

where ∇_T^2 is the transverse Laplace operator, $i = \sqrt{-1}$ is the imaginary unit, and $\partial/\partial z$ denotes the paraxial derivative with respect to the propagation direction z. The quantities $\tilde{u}$ and k are defined as before [6].

One solution of the paraxial Helmholtz equation is the paraboloidal wave

$$\tilde{u}(\rho, z) = \frac{A_1}{z} \exp\left(-ik\frac{\rho^2}{2z}\right), \tag{2.4}$$

where A_1 is a constant complex amplitude, z is the propagation distance, and

$$\rho^2 = x^2 + y^2 \tag{2.5}$$

defines the radial coordinate in the transverse plane. Here, x and y are the transverse Cartesian coordinates [6].

Using the complex beam parameter

$$q(z) = z + iz_R, \tag{2.6}$$

where z_R is the Rayleigh length, the Gaussian beam solution can be written as

$$\tilde{u}(\rho, z) = \frac{A_1}{q(z)} \exp\left[-ik\frac{\rho^2}{2q(z)}\right] \tag{2.7}$$

[6].

The complete complex electric field amplitude of a Gaussian beam is then given by

$$\tilde{E}(\rho, z) = A_0 \frac{w_0}{w(z)} \exp\left[-\frac{\rho^2}{w^2(z)}\right] \exp\left[-ikz - ik\frac{\rho^2}{2R(z)} + i\zeta(z)\right], \tag{2.8}$$

where A_0 is the field amplitude at the beam waist, w_0 is the beam waist radius, $w(z)$ is the beam radius at the position z, $R(z)$ is the radius of curvature of the wavefront, and $\zeta(z)$ is $\boxed{\zeta(z) = \arctan\left(\frac{z}{z_R}\right)}$ [8].

Having established the field distribution of a Gaussian beam, the next step is to examine the parameters that characterize its propagation. In particular, the beam waist and beam divergence provide a quantitative description of the beam size and its evolution along the propagation direction.

2.1.2 Beam Waist and Divergence

The beam radius $w(z)$ reaches its minimum value at the beam waist, which is located in the transverse plane defined by $z = 0$. The minimum beam radius at this position is denoted by w_0 [9].

The propagation of the Gaussian beam radius along the z-direction is described by

$$w(z) = w_0 \left[1 + \left(\frac{z}{z_R}\right)^2\right]^{1/2} \tag{2.9}$$

where z_R is the Rayleigh range, defined as the distance from the beam waist at which the beam radius has increased by a factor of $\sqrt{2}$ [8].

Far from the beam waist, the beam radius increases approximately linearly with the propagation distance, forming a divergence cone. The half-angle beam divergence is given by

$$\theta_0 = \frac{\lambda}{\pi w_0}, \tag{2.10}$$

where λ is the wavelength of the laser beam [9].

The Gaussian beam described in the previous sections represents an idealized beam and provides the theoretical lower limit for beam divergence at a given beam waist. In practice, however, real laser beams often deviate from this ideal behavior. To quantify the extent of these deviations and compare the quality of different laser sources, the beam quality factor M^2 is introduced.

2.1.3 Beam Quality Factor

Different resonator geometries support different transverse solutions of the wave equation and therefore different transverse laser modes. The fundamental transverse electromagnetic mode is the TEM_{00} mode, which exhibits a Gaussian intensity distribution [10]. By shifting the waist position to z_0, the propagation of the beam radius from Equation (2.9) can be rewritten as [11]

$$w^2(z) = w_0^2 \left[1 + \left(\frac{z - z_0}{z_R}\right)^2\right]. \tag{2.11}$$

For an ideal Gaussian beam, the intensity distribution is given by

$$I(r) = I_0 \exp\left(-\frac{2r^2}{w^2}\right), \tag{2.12}$$

where $I(r)$ is the optical intensity at the radial distance r from the beam center, I_0 is the peak intensity at the beam center ($r = 0$), and w is the beam radius, defined as the radius at which the intensity has decreased to $1/e^2$ of its maximum value [11].

Anthony Siegman generalized this formalism for arbitrary real laser beams by introducing beam widths based on the statistical standard deviation of the intensity distribution:

$$w_x(z) = 2\sigma_x(z), \tag{2.13}$$

where $w_x(z)$ is the beam radius in the x-direction at the propagation distance z, and $\sigma_x(z)$ is the standard deviation of the intensity distribution in the x-direction [11].

The propagation of real, non-ideal laser beams can then be described by the M^2-formalism:

$$w_x^2(z) = w_{0x}^2 + M_x^4 \left(\frac{\lambda}{\pi w_{0x}}\right)^2 (z - z_{0x})^2, \tag{2.14}$$

where w_{0x} is the minimum beam radius (beam waist) in the x-direction, M_x^2 is the beam quality factor in the x-direction, and z_{0x} is the position of the beam waist [11].

The beam quality factor satisfies

$$M^2 \geq 1, \tag{2.15}$$

where $M^2 = 1$ corresponds to an ideal diffraction-limited Gaussian beam, while larger values indicate reduced beam quality due to higher-order modes or optical distortions [11].

The divergence of a real beam is therefore given by

$$\theta = M^2 \frac{\lambda}{\pi w_0}, \tag{2.16}$$

where θ is the far-field divergence half-angle [11]. This equation shows that beams with $M^2 > 1$ diverge more strongly than an ideal Gaussian beam.

The beam quality factor M^2 therefore provides an important criterion for evaluating laser sources.

After the basic properties of laser beams have been introduced, the next step is to describe how such beams can be used for displacement measurements. For this reason, the following section explains the principle of interferometry.

2.2 Systems for Interferometry

Interferometry is a general optical measurement technique that exploits the interference of coherent light waves to determine quantities such as displacement with high precision. Numerous interferometer configurations have been developed for different applications, each based on the same fundamental principle of combining two or more light beams to produce an interference pattern. One of the simplest and most widely used configurations is the Michelson interferometer [12].

2.2.1 Michelson Interferometer

A Michelson interferometer consists of a beam splitter and two mirrors. A coherent light beam enters the interferometer and a beam splitter divides it into two beams of identical intensity that travel along different optical paths. After being reflected by the mirrors, the beams recombine [13]. The setup is shown in Figure 2.1.

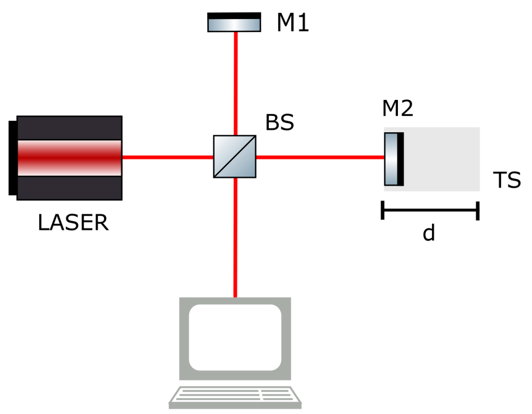

Figure 2.1: Schematic representation of a Michelson interferometer. The incoming beam is split at the beam splitter, denoted by BS, into two beams that propagate along the two interferometer arms. After reflection at mirrors M1 and M2, the beams recombine at the beam splitter and their interference can be detected at the output. The mirror M2 creates an optical path difference by moving a distance d on a Translation Stage TS.

The electric field of the incoming plane wave can be written as

$$E_l = A_l \cos(\omega t - kz), \tag{2.17}$$

where A_l is the field amplitude, ω is the angular frequency, and t is the time [13].

The electric fields of the two partial waves E_1 and E_2 after propagation through the interferometer are given by

$$E_1 = \sqrt{RT} A_l \cos(\omega t + \varphi_1) \tag{2.18}$$

and

$$E_2 = \sqrt{RT} A_l \cos(\omega t + \varphi_2), \tag{2.19}$$

where R and T denote the intensity reflection and transmission coefficients of an ideal, lossless beam splitter ($R + T = 1$), and φ_1 and φ_2 are the optical phases accumulated in the two interferometer arms [13].

The detected intensity results from the superposition of the two electric fields:

$$\begin{aligned} I_T &= c\varepsilon_0 (E_1 + E_2)^2 \\ &= c\varepsilon_0 RT A_l^2 \left[\cos(\omega t + \varphi_1) + \cos(\omega t + \varphi_2)\right]^2, \end{aligned} \tag{2.20}$$

where c is the speed of light and ε_0 is the vacuum permittivity [13].

Since the detector averages over many optical periods, the time-averaged intensity $\bar{I}_T = \langle I_T \rangle$ is measured. Using the trigonometric identity for the sum of cosines and taking the time average of Eq. (2.20) yields

$$\bar{I}_T = c\varepsilon_0 RT A_l^2 (1 + \cos \Delta\varphi). \tag{2.21}$$

By defining the time-averaged intensity of the incident laser beam as

$$I_0 = \frac{1}{2} c\varepsilon_0 A_l^2, \tag{2.22}$$

the final expression for the detected time-averaged intensity becomes

$$\bar{I}_T = 2RT I_0 (1 + \cos \Delta\varphi), \tag{2.23}$$

with the phase difference

$$\Delta\varphi = \varphi_1 - \varphi_2 = \frac{2\pi}{\lambda} \Delta s, \tag{2.24}$$

where Δs is the optical path difference between the two interferometer arms [2].

2.2.1.1 Interference

As already noted in Eq. (2.24), the optical path difference determines the phase difference between the two interfering waves. Interference describes the superposition of coherent electromagnetic waves. Constructive interference occurs when the optical path difference is an integer multiple of the wavelength,

$$\Delta s = m\lambda, \tag{2.25}$$

which corresponds to a phase difference of

$$\Delta\varphi = 2\pi m, \tag{2.26}$$

where $m \in \mathbb{Z}$. Conversely, destructive interference occurs for optical path differences of

$$\Delta s = \left(m + \frac{1}{2}\right)\lambda, \tag{2.27}$$

corresponding to a phase difference of

$$\Delta\varphi = (2m+1)\pi. \tag{2.28}$$

Therefore, very small displacements can be detected by counting interference fringes [14].

2.2.1.2 Fringe Counting

The Michelson interferometer was chosen because it is a simple configuration for displacement measurements [2]. One method of evaluating the interference signal from it is fringe counting. Since this method is applied in this work, it is briefly introduced in the following.

In the Michelson interferometer, the movement of one mirror changes the optical path difference between the two interferometer arms. Whenever the optical path difference changes by one wavelength (λ), one complete interference fringe passes the detector. Since the light in that arm travels the added distance twice, once towards the mirror and once back, a mechanical mirror displacement d produces an optical path difference of

$$\Delta s = 2d. \tag{2.29}$$

After N counted fringes, the optical path difference has therefore changed by $\Delta s = N\lambda$. Combining this with Eq. (2.29), the stage displacement d of the moving mirror is related to the number of counted fringes N by

$$d = \frac{N\lambda}{2} \tag{2.30}$$

[2].

2.2.2 Quadrature Homodyne Interferometry

A conventional Michelson interferometer like explained above is a homodyne interferometer, meaning that both interfering beams originate from the same laser source and therefore possess the same optical frequency [15].

While a conventional homodyne interferometer can measure displacement with high precision, it cannot determine the direction of motion. This limitation arises because the detected intensity depends on the cosine of the phase difference, which is an even function:

$$\cos(\Delta\varphi) = \cos(-\Delta\varphi). \tag{2.31}$$

As a result, identical intensity values are obtained for positive and negative displacements, making it impossible to distinguish whether the translation stage is moving forward or backward [16].

To overcome this ambiguity, an alternative homodyne interferometry technique, namely quadrature homodyne interferometry, was developed. In this technique, two interference signals with a phase offset of 90° are generated [16]. The resulting detector signals can be written as

$$S_1 = A\cos(\Delta\varphi), \tag{2.32}$$

and

$$S_2 = A\sin(\Delta\varphi), \tag{2.33}$$

where A is the signal amplitude. These signals are commonly referred to as quadrature signals because they are shifted by one quarter of a period relative to each other [16]. In the setup, shown in Figure 2.2, the simultaneous detection of the s- and p-polarized components provides directional information about the phase shift. In contrast, a conventional interferometer typically measures only one polarization component, making it difficult to distinguish the sign of the phase change.

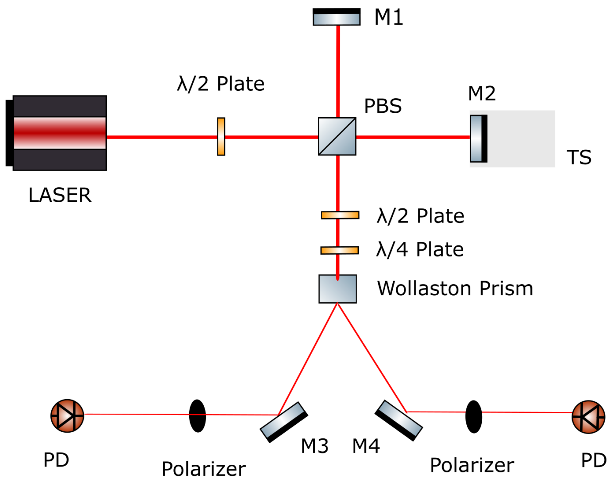

Figure 2.2: Setup for quadrature homodyne interferometry. This configuration visualizes the principle of quadrature signal generation. PBS denotes a polarizing beam splitter, M1 – M4 denote mirrors and PD a photodiode.

The two signals can be represented as coordinates in the (S_1, S_2) plane, as shown in Figure 2.3. Ideally, the resulting trajectory forms a circle as the stage moves. The phase can then be reconstructed continuously using

$$\Delta\varphi = \text{atan2}(S_2, S_1), \tag{2.34}$$

which uniquely determines the phase over the full 2π range [17].

Unlike a conventional homodyne interferometer, the direction of motion can now be determined from the temporal evolution of the reconstructed phase. As the translation stage moves in the positive direction, the phase increases continuously. Conversely, motion in the opposite direction causes the phase to decrease [16]. The direction is therefore obtained from the sign of the phase change between consecutive measurements,

$$\frac{d\Delta\varphi}{dt} > 0 \quad \Rightarrow \quad \text{forward motion,} \tag{2.35}$$

$$\frac{d\Delta\varphi}{dt} < 0 \quad \Rightarrow \quad \text{backward motion.} \tag{2.36}$$

This can also be interpreted geometrically. In the (S_1, S_2) plane, positive motion causes the measured signal vector to rotate counterclockwise, whereas negative motion results in

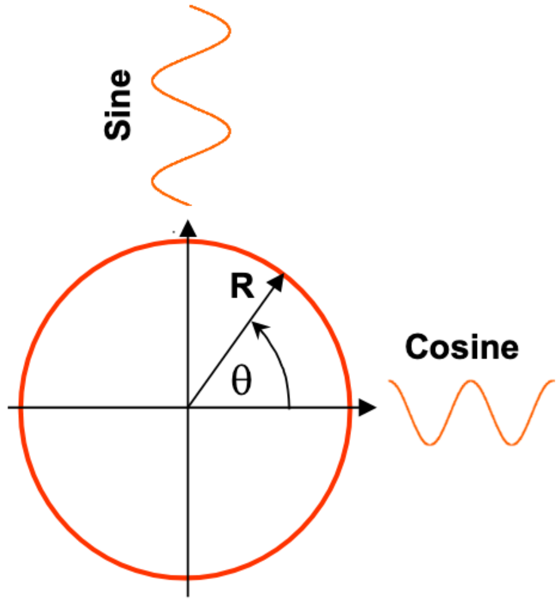

Figure 2.3: Principle of direction-sensitive quadrature signal evaluation in homodyne interferometry. The phase relation between two quadrature signals enables the direction of motion to be determined. Obtained from [17].

clockwise rotation. By tracking the phase continuously, both displacement and direction can therefore be determined unambiguously.

The displacement is obtained from the measured phase according to

$$d = \frac{\lambda}{4\pi}\Delta\varphi. \tag{2.37}$$

[18].

Since the phase can be tracked continuously, quadrature homodyne interferometry can measure displacements which are much smaller than the laser wavelength [19]. It also provides information about the direction of movement, making it possible to track the position reliably over long distances. For these reasons, a quadrature homodyne interferometer was chosen as an extension for the displacement measurement system developed in this thesis.

To realize the expanded capabilities of quadrature homodyne interferometry compared to a conventional Michelson interferometer, additional optical elements are required. Specifically, polarization-controlling components are essential for generating the quadrature signals. These components and their operating principles are introduced in the following.

2.2.2.1 Optical Components for Quadrature Homodyne Interferometry

For completeness, the most important optical components used in the quadrature homodyne interferometer are briefly introduced. Specifically, a $\lambda/2$ wave plate, a $\lambda/4$ wave plate and a Wollaston prism are the main components.

Half-Wave Plate A half-wave plate is a special type of wave retarder. In general, a wave retarder is a birefringent optical element with two principal axes, usually referred to as the fast and slow axes. The electric field of an incident beam can be resolved into components along these two axes. Since the refractive indices differ for the fast and slow axes, the components propagate with different phase velocities, resulting in a relative phase delay between them. [20]

For a half-wave plate, this phase retardance is

$$\Gamma = \pi. \tag{2.38}$$

This means that one polarization component is delayed by half an optical period with respect to the orthogonal component. For linearly polarized input light, the output remains linearly polarized, but the plane of polarization is rotated. If the incident polarization forms an angle θ with one of the principal axes of the half-wave plate, the polarization direction after the plate is rotated by 2θ with respect to the incident direction [21].

In the quadrature homodyne interferometer, this property is used to adjust the polarization angle of the laser beam. By rotating the polarization by 45° with respect to the axes of a polarizing beam splitter, the optical power can be divided into two equal polarization components.

Quarter-Wave Plate A quarter-wave plate operates in the same way as a half-wave plate but introduces a phase retardance of

$$\Gamma = \frac{\pi}{2}. \tag{2.39}$$

Consequently, one polarization component is delayed by one quarter of an optical period relative to the orthogonal component. If linearly polarized light enters the wave plate at an angle of 45° to the principal axes, the output becomes circularly polarized. Conversely, circularly polarized light is converted into linearly polarized light [22].

Wollaston Prism A Wollaston prism, shown in Figure 2.4, is a polarizing beam splitter that separates an incoming beam into two linearly polarized output beams. It consists of two birefringent wedge prisms, commonly made from a material such as calcite. Although a magnesium fluoride (MgF_2) Wollaston prism was used in this work due to availability, this choice of material does not affect the operating principle. The two wedges are joined together with their optical axes oriented perpendicular to each other [22]. Because birefringent materials exhibit different refractive indices for different polarization directions, the ordinary and extraordinary polarization components experience different refraction angles. This difference in refraction arises because the angle of refraction depends on the refractive indices of the media, as described by Snell's law,

$$n_1 \sin\theta_1 = n_2 \sin\theta_2, \tag{2.40}$$

where n_1 and n_2 are the refractive indices of the first and second medium, respectively, and θ_1 and θ_2 are the angles of incidence and refraction, measured with respect to the surface normal [23].

In the first prism, one polarization component propagates as the ordinary ray while the orthogonal component propagates as the extraordinary ray. In the second prism, whose optical axis is rotated by 90°, these roles are interchanged. As a result, the two polarization components are refracted in opposite directions at the internal interface and leave the prism as two spatially separated beams. The use of two prisms increases the angular separation and makes the two output beams easier to use experimentally.

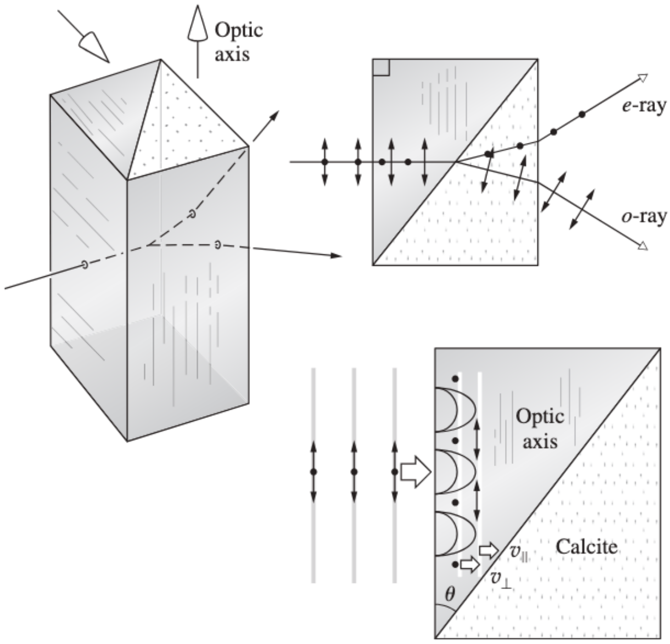

Figure 2.4: Schematic representation of a Wollaston prism. The two cemented birefringent prisms have perpendicular optical axes, causing orthogonally polarized components to be refracted in different directions. Obtained from [22].

Other components Other components include a conventional 50/50 beam splitter and a polarizing beam splitter, whose basic functions are only mentioned briefly here. A conventional 50/50 beam splitter divides the incoming beam into two beams of equal intensity, while a polarizing beam splitter separates the beam into two orthogonal polarization components.

After introducing the different interferometric configurations, the next step is to consider how such a setup can be realized experimentally. One of the most important components is the laser source. Since different laser technologies have different advantages and limitations, the following section introduces two laser types that are especially relevant for interferometric applications.

2.3 Laser Sources for Interferometry

Traditionally, helium-neon (HeNe) lasers are widely used in interferometry [24]. Semiconductor diode lasers, however, have become increasingly attractive due to their compact design and high efficiency [25].

In this work, several HeNe lasers and one semiconductor diode laser were investigated as potential laser sources for the interferometric calibration system. The following sections summarize the operating principles and characteristic properties of both laser types.

2.3.1 Helium-Neon (HeNe) Lasers

The Helium-Neon (HeNe) laser is one of the most widely used gas lasers in optics and interferometry. HeNe lasers emit coherent red light at a wavelength of 632.8 nm and operate in continuous-wave mode [26]. The emitted laser beam exhibits a narrow spectral bandwidth, high temporal and spatial coherence as well as low beam divergence [27]. High coherence is essential because stable interference fringes can only be observed, if the phase relation between the interfering waves remains constant over time [28]. Due to these properties, HeNe lasers are particularly suitable for interferometric measurements and precision optical experiments [27].

2.3.1.1 Operating Principle

A schematic setup of a typical HeNe laser is shown in Figure 2.5. The laser consists of a gas discharge tube filled with a mixture of helium and neon gas [29].

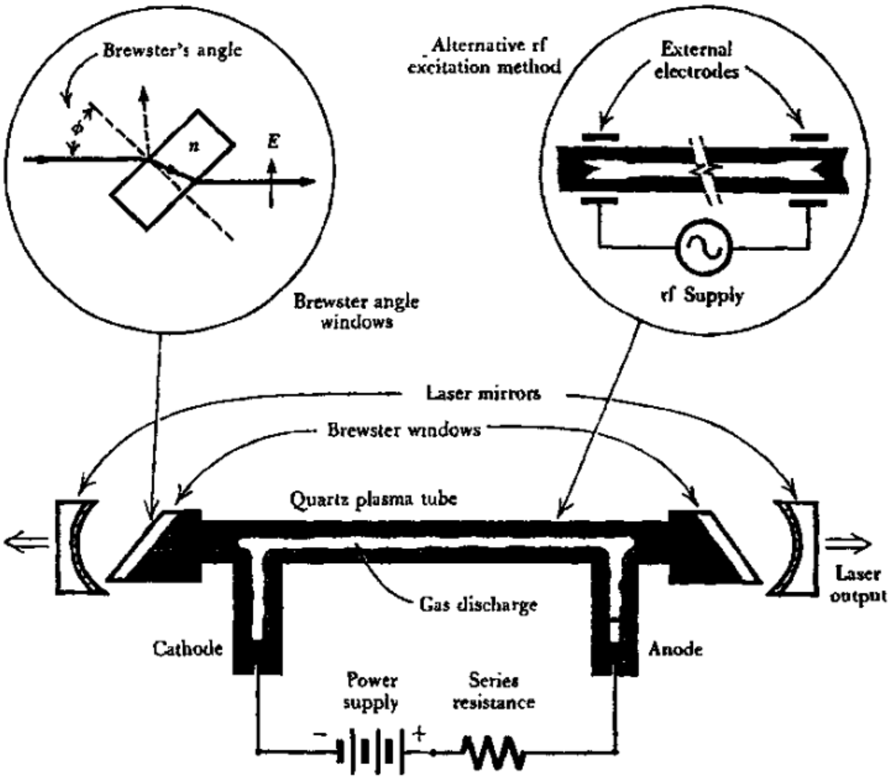

Figure 2.5: Schematic setup of a HeNe laser (obtained from [30]).

The laser process in a HeNe laser begins with an electrical gas discharge inside a mixture of helium and neon gas. A short high-voltage pulse excites free electrons within the discharge tube. These electrons collide predominantly with helium atoms and excite them into metastable energy states [31].

The energies of the metastable helium states match specific excited energy levels of neon atoms. During collisions between helium and neon atoms, the excitation energy can therefore be transferred efficiently from helium to neon. As a consequence, neon atoms are excited into higher energy states without requiring direct excitation [31].

The continuous transfer of energy from helium to neon leads to the formation of a population inversion in the neon atoms. In this condition, more neon atoms occupy excited states than lower-energy states, which is a necessary requirement for laser amplification. Neon therefore forms the active gain medium of the laser, whereas helium mainly serves as the pumping medium that enables the excitation process [31].

Figure 2.6 illustrates the simplified energy level scheme of the HeNe laser.

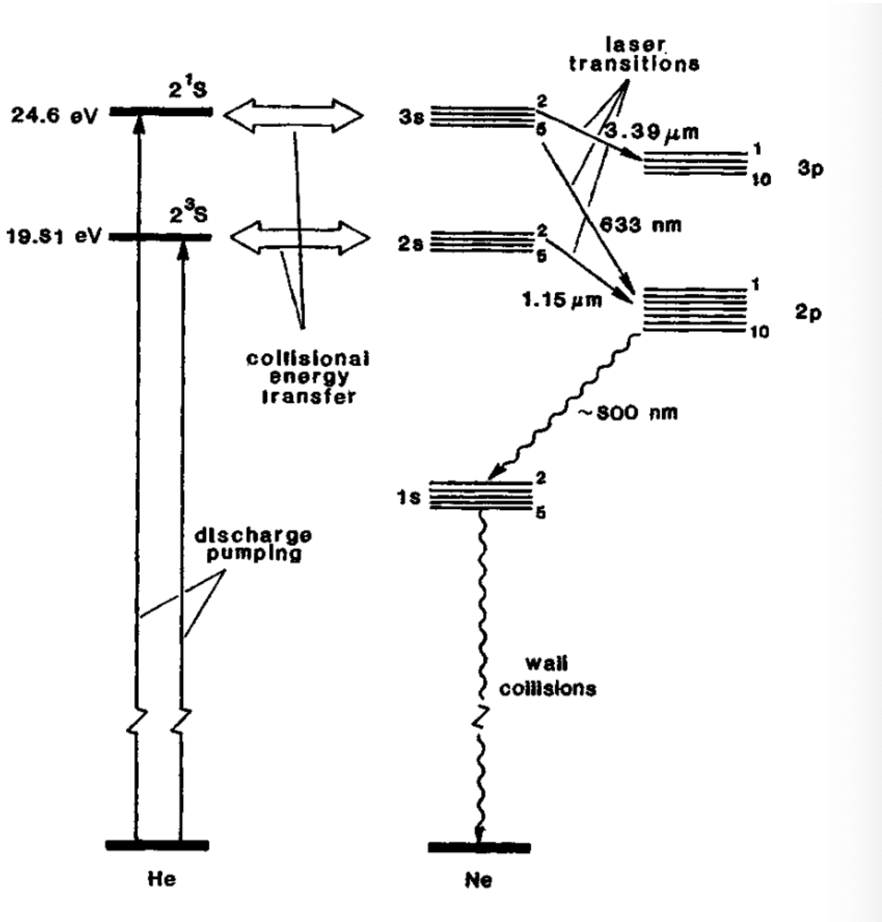

Figure 2.6: Simplified energy level diagram of the HeNe laser (obtained from [32]).

Laser emission occurs through stimulated emission in neon atoms. Excited neon atoms transition from the upper laser level to a lower energy level and emit photons with a wavelength of 632.8 nm. These photons are reflected repeatedly between two mirrors forming the optical resonator. One mirror has a very high reflectivity, while the second mirror is partially transmitting. The repeated reflections amplify the optical field inside the resonator, and a fraction of the amplified radiation exits through the partially transmitting mirror as the visible laser beam [31].

2.3.2 Semiconductor Diode Lasers

Semiconductor diode lasers are compact and efficient laser sources. In contrast to gas lasers such as HeNe lasers, laser generation in diode lasers is based on charge carrier recombination inside a semiconductor pn-junction [33].

Diode lasers offer several practical advantages including compact size, low electrical power consumption, high efficiency, and comparatively low manufacturing costs [25].

2.3.2.1 Operating Principle

A semiconductor diode laser, shown in Figure 2.7, consists of two differently doped semiconductor regions forming a pn-junction. One semiconductor region contains an excess of electrons and is referred to as the n-type material, while the other region contains an excess of holes and is referred to as the p-type material [25].

When a forward bias voltage is applied across the pn-junction, electrons from the n-type region and holes from the p-type region are injected into the junction region [33]. In this active region, electrons and holes recombine and release energy in the form of photons [25].

The emitted photons stimulate further recombination events, resulting in stimulated emission. The resonator is typically formed by partially reflective semiconductor surfaces located at both ends of the laser chip [25].

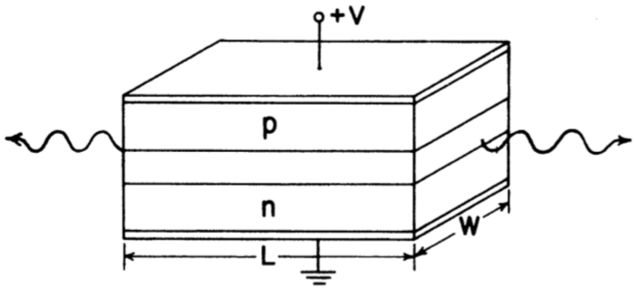

Figure 2.7: Schematic structure of a semiconductor diode laser. The active region is located at the pn-junction, where injected electrons and holes recombine and generate optical radiation. Obtained from [27].

Compared to HeNe lasers, diode lasers often exhibit larger beam divergence [34] and stronger temperature dependence [35]. However, their compactness, efficiency, and practical stability characteristics make them attractive candidates for interferometric measurement systems [36].

After introducing the laser sources, the translation stage used in the experimental setup must also be considered. Since the interferometer is designed to measure and improve the positioning accuracy of the stage, its operating principle and limitations are discussed in the following.

2.4 Motorized Translation Stage

A motorized precision translation stage is used in the interferometer setup to generate controlled linear displacements of the mirror.

2.4.1 Operating Principle

The motorized translation stage used in this work (Thorlabs LTS300C/M) operates according to this fundamental principle: Linear motion is generated by converting the rotational motion of an electric motor into translational motion using a lead screw. The moving platform is guided by rolling-element linear bearings, which provide smooth motion with high stiffness and low friction [37, 38].

Although motorized translation stages provide highly repeatable and accurate positioning over long travel distances, their performance is inherently limited by both mechanical and environmental influences. These effects prevent the commanded position from exactly matching the actual position of the moving platform. The positioning accuracy is affected by mechanical tolerances, such as lead screw errors and bearing imperfections, as well as by thermal effects. Temperature variations cause thermal expansion of structural components, guide rails, and the drive mechanism, resulting in positioning errors, particularly over long travel distances [39].

Due to these mechanical and thermal limitations, the commanded stage position cannot be assumed to represent the true displacement of the mirror with nanometer accuracy. Therefore, an independent interferometric measurement system is used in this work to determine the actual displacement with higher precision.

CHAPTER 3

LASER CHARACTERIZATION AND SELECTION

3.1 Requirements for the Laser Source

The performance of an interferometric measurement system strongly depends on the properties of the employed laser source. Since the displacement measurement in an interferometer is directly related to the laser wavelength, the laser must provide stable emission characteristics and sufficient beam quality. In addition, intensity fluctuations and beam distortions can reduce the visibility of the interference fringes and therefore decrease the measurement accuracy [28].

A total of six different laser sources were characterized experimentally. The characterization included wavelength measurements using a spectrometer, beam profile measurements for the determination of the beam quality factor M^2, and intensity stability measurements using a power meter.

3.2 Wavelength Measurements

The wavelengths of the investigated laser sources were measured using a spectrometer. The laser beam was first redirected using a mirror and then coupled into an optical fiber connected to the spectrometer using a convex CaF_2 lens with a focal length of 50 mm. The setup is shown in Figure 3.1.

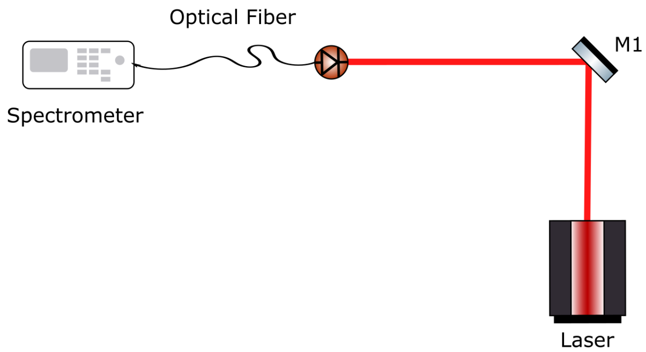

Figure 3.1: Experimental setup and schematic representation of the wavelength measurements with the spectrometer. In the schematic, M1 denotes the mirror.

The wavelengths of the laser sources were measured using an Ocean Optics SR6VN500-10 spectrometer covering the spectral range from 300 nm to 1100 nm. Due to the high optical power of the laser sources, the detector signal exceeded the dynamic range of the OceanView acquisition software, leading to saturation. Therefore absorptive neutral density filters were inserted into the beam path between the mirror and the spectrometer input.

Additionally, for every measurement a background spectrum without laser illumination was recorded. This background signal was subtracted from the measured spectra in order to reduce detector noise.

The measured peak wavelengths of the investigated laser sources are summarized in Table 3.1. The corresponding background-corrected spectra are shown in Figure 3.2.

Table 3.1: Measured peak wavelengths of the investigated laser sources.

Laser Source	Peak Wavelength (nm)
Thorlabs CPS780S	787.32
Uniphase 1023P	631.81
Uniphase 1103P 1177761	631.81
Uniphase 1103P 1108380	632.01
Uniphase 1122P	632.21
Uniphase 1507P	634.82

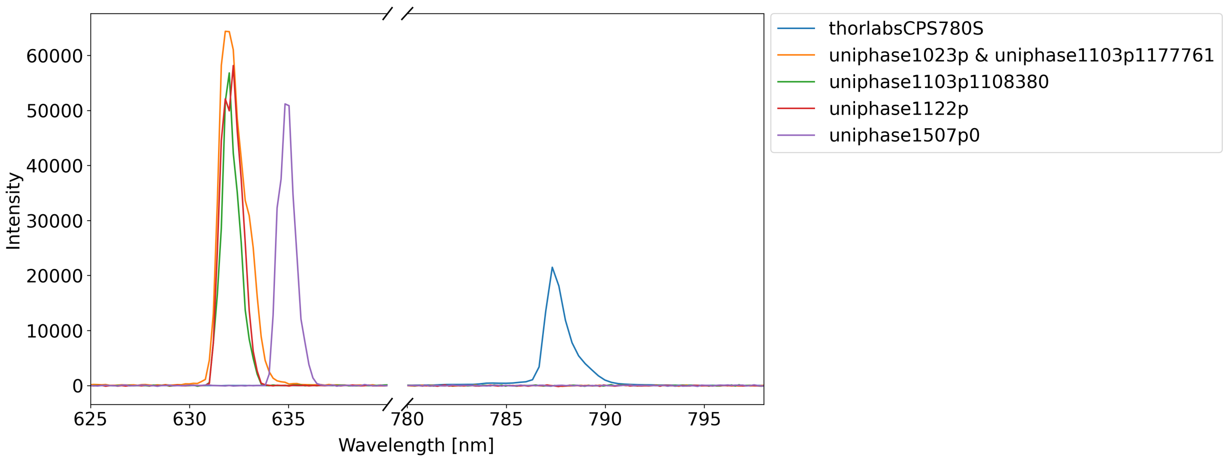

Figure 3.2: Background-corrected spectra of the investigated laser sources used to determine the peak emission wavelengths. The spectra of the Uniphase 1023P and Uniphase 1103P (1177761) were identical, apart from severe detector saturation in the latter. Therefore, the unsaturated spectrum is used to represent both sources. Although minor saturation, clearly distinguishable from the severe saturation of the 1103P (1177761), also occurred for some other spectra (such as the Uniphase 1507P), the peak emission wavelengths could still be determined accurately in all cases.

The Uniphase HeNe laser sources emitted wavelengths close to the expected wavelength of approximately 632.8 nm. The measured deviations are likely caused by differences in the laser cavity, the gas composition, and the gas pressure but could also originate from spectrometer-calibration or resolution. The Thorlabs Diode Laser emitted a wavelength of 787.3 nm.

3.3 Beam Profile Measurements and Determination of Beam Quality Factor

As introduced in Chapter 2, another key parameter for characterizing laser beams is the beam quality factor M^2. In this work, M^2 was determined using a CinCam beam profiling camera. For the measurements, the laser beam was redirected by a mirror and focused using a lens with a focal length of 75 mm. The experimental setup is shown in Figure 3.3.

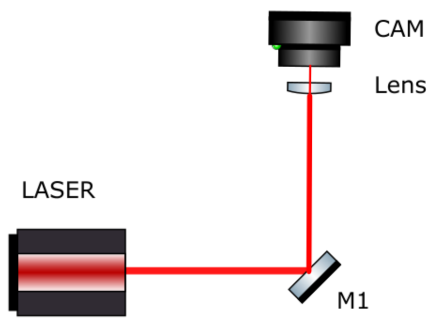

Figure 3.3: Experimental setup and schematic representation of the beam profile measurements with the CinCam beam profiling camera. In the schematic, M1 denotes the mirror and CAM denotes the CinCam beam profiling camera.

Since the optical power of several laser sources exceeded the permissible detector range of the camera sensor, absorptive neutral density filters were inserted into the beam path to prevent detector saturation.

3.3.1 Beam Diameter Measurements

The beam profile measurements were performed using the RayCi software. For each laser source, the beam diameter was measured along the propagation direction in steps of 1 mm. The beam diameters were determined by the RayCi software separately for both transverse beam axes using the D4σ method in accordance with ISO 11146 [40].

The D4σ method defines the beam diameter based on the second moments of the measured intensity distribution and is particularly suitable for the characterization of non-ideal laser beams [41]. The resulting beam diameters were recorded as a function of the propagation distance and subsequently exported for further analysis.

3.3.2 Determination of the Beam Quality Factor

While the RayCi software provides beam diameter measurements, it does not directly calculate the beam quality factor M^2. Therefore, the exported D4σ beam diameters were evaluated using a dedicated analysis program provided by Dr. Sergio Revuelta.

The program performs a nonlinear least-squares fit of the measured beam radii to the propagation model of a real Gaussian beam according to ISO 11146. The beam quality factor is defined as

$$M^2 = \frac{\pi w_0 \theta}{\lambda}, \tag{3.1}$$

where w_0 is the beam waist radius, θ is the far-field divergence angle, and λ is the laser wavelength [42].

Using the measured beam radii as a function of propagation distance, the fitting software determines the beam waist radius w_0, the waist position z_0, the divergence angle θ, the Rayleigh range z_R, and the corresponding beam quality factor M^2 independently for both transverse beam axes. The fitted propagation curves were compared with the measured data to assess the fit quality and the applicability of the Gaussian beam propagation model. To ensure reliable M^2 determination, ISO 11146 requires measurement points both near the beam waist and in the far field.

Figure 3.4 shows the combined measured beam radii of all beams and the corresponding fitted propagation curves. The minimum of each fitted curve indicates the beam waist position, while the slope in the far field determines the beam divergence. Again, all Uniphase HeNe lasers exhibited very similar propagation curves and M^2 values, whereas the Thorlabs diode laser showed a significantly higher M^2 value, indicating poorer beam quality.

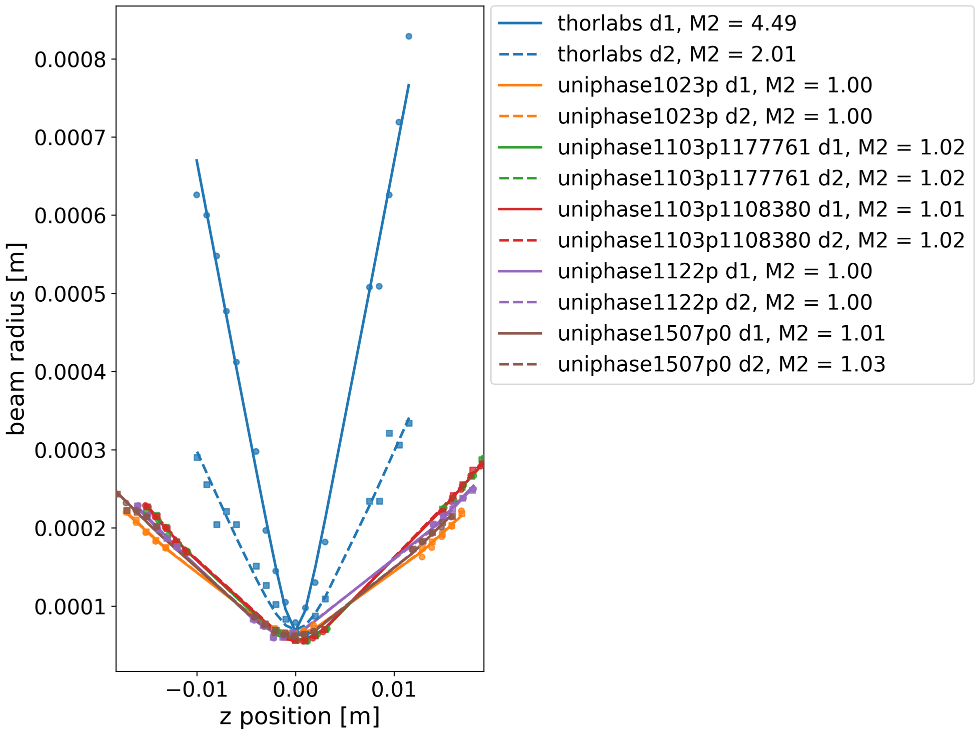

Figure 3.4: Measured beam radii and fitted beam propagation curves. The beam propagation model was fitted independently for both transverse beam axes. Discrete data points represent the measured values, while lines indicate the corresponding fits.

The resulting beam propagation parameters for all investigated laser sources are summarized in Table 3.2.

Table 3.2: Measured beam propagation parameters of the investigated laser sources.

Laser Source	M_x^2	M_y^2	Waist Radius x (μm)	Waist Radius y (μm)
Thorlabs CPS780S	4.49	2.01	67.7	69.5
Uniphase 1023P	1.00	1.00	66.2	65.8
Uniphase 1103P 1177761	1.02	1.02	56.5	56.6
Uniphase 1103P 1108380	1.01	1.02	56.0	55.6
Uniphase 1122P	1.00	1.00	59.0	59.3
Uniphase 1507P	1.01	1.03	62.2	63.9

3.4 Intensity Stability Measurements

The temporal intensity stability of the laser sources was investigated using an Ophir Vega power meter. The laser beam was redirected using a mirror and directed onto the power meter detector. The experimental setup is shown in Figure 3.5.

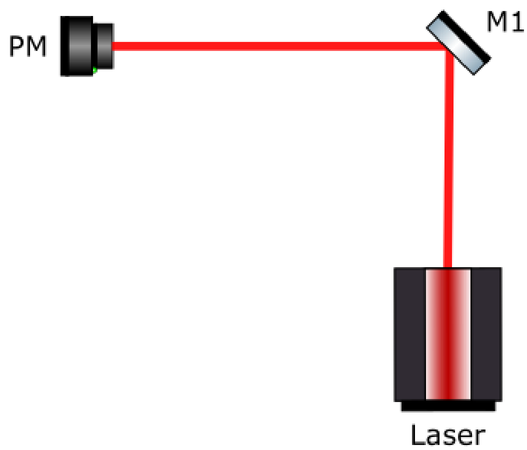

Figure 3.5: Experimental setup and schematic representation of the intensity stability measurements with the Ophir Vega power meter. In the schematic, M1 denotes the mirror and PM denotes the power meter.

For each laser source, the optical power was recorded over a time interval of approximately 600 s. The measured signals were analyzed with respect to their mean optical power, standard deviation, and signal-to-noise ratio (SNR). Figure 3.6 shows the recorded power measurements of all investigated laser sources.

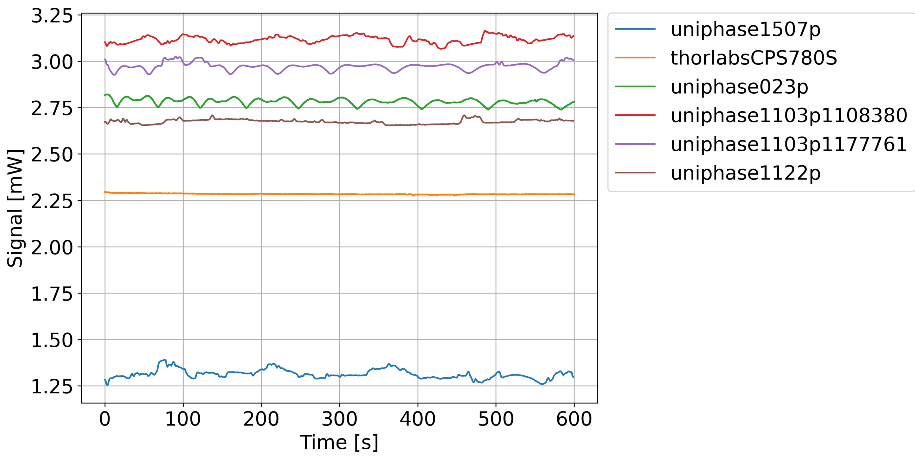

Figure 3.6: Comparison of the measured laser intensities over time.

The calculated SNR values are summarized in Table 3.3.

The Thorlabs laser source exhibited the highest signal-to-noise ratio and the smallest intensity fluctuations over time. All of the Uniphase laser sources showed periodic fluctuations and increased noise contributions, which negatively affect interferometric fringe stability.

Table 3.3: Measured signal-to-noise ratios of the investigated laser sources.

Laser Source	SNR
Uniphase 1507P	53.11
Thorlabs CPS780S	836.51
Uniphase 1023P	170.13
Uniphase 1103P 1108380	155.98
Uniphase 1103P 1177761	146.00
Uniphase 1122P	231.11

3.5 Selection of the Final Laser Source

Based on the wavelength measurements, beam profile analysis, and intensity stability measurements, the investigated laser sources were evaluated with respect to their suitability for interferometric displacement measurements.

Among the tested laser sources, the Thorlabs laser exhibited the highest intensity stability and the best signal-to-noise ratio. Although its operating wavelength differs from that of the investigated HeNe lasers, this does not constitute a disadvantage for the intended interferometric measurements, as the measurement principle itself is independent of the absolute wavelength, provided that the wavelength is accurately known.

The beam quality analysis revealed that the Thorlabs laser exhibited the highest M^2 value among the investigated sources, indicating a greater deviation from an ideal Gaussian beam. However, for the Michelson interferometer developed in this work, intensity stability and signal quality were found to be more critical than achieving a near-ideal Gaussian beam profile, since the fringe detection algorithm analyzes only the central part of the beam, making the overall beam shape of secondary importance. In contrast, intensity fluctuations may cause incorrect fringe detection, directly affecting the accuracy of the displacement measurement. Consequently, the superior stability characteristics of the Thorlabs laser outweighed its reduced beam quality, and it was therefore selected as the light source for all subsequent interferometric measurements.

CHAPTER 4

SOFTWARE DEVELOPMENT

The subsequent chapter focuses on the software developed to evaluate the interferometric signals. The primary objective of the software was to provide an independent measurement of stage displacement based on interferometric fringe counting and directly comparing this value with the displacement reported by the translation stage controller. For this purpose, three software implementations were developed: one for camera-based fringe counting, one for photodiode-based fringe counting, and one for evaluating the quadrature homodyne interferometer.

4.1 Software for Fringe detection with the Camera

Several functional requirements were identified before development began.

First, the software had to continuously acquire images from the camera while maintaining a responsive user interface. Second, it had to reliably detect and count interference fringes in real time despite fluctuations in illumination and environmental noise. Third, direct control of the electronic translation stage was required, including both relative and absolute positioning commands.

Finally, the measured interferometric displacement had to be continuously compared with the displacement reported by the stage controller in order to evaluate positioning accuracy.

The software was implemented in Python and organized into three main components: a graphical user interface, a camera handling module, and a translation stage control module.

4.1.1 Graphical User Interface

The graphical user interface was implemented using the `CustomTkinter` library. It provides all user interaction functions, including starting and stopping measurements, entering the laser wavelength, controlling stage motion and displaying measurement results.

A screenshot of the user interface implementation is shown in Figure 4.1.

The interface displays the current fringe intensity, accumulated fringe count, interferometrically measured displacement, optical time delay, stage position, and the difference between commanded and measured displacement. A live image of the camera is also displayed to assist with alignment and monitoring.

4.1.2 Camera Handler

Communication with the Thorlabs camera was implemented in a dedicated `CameraHandler` module. This module is responsible for loading the required camera drivers, acquiring images, extracting the fringe intensity from each frame, and providing image data to the graphical user interface.

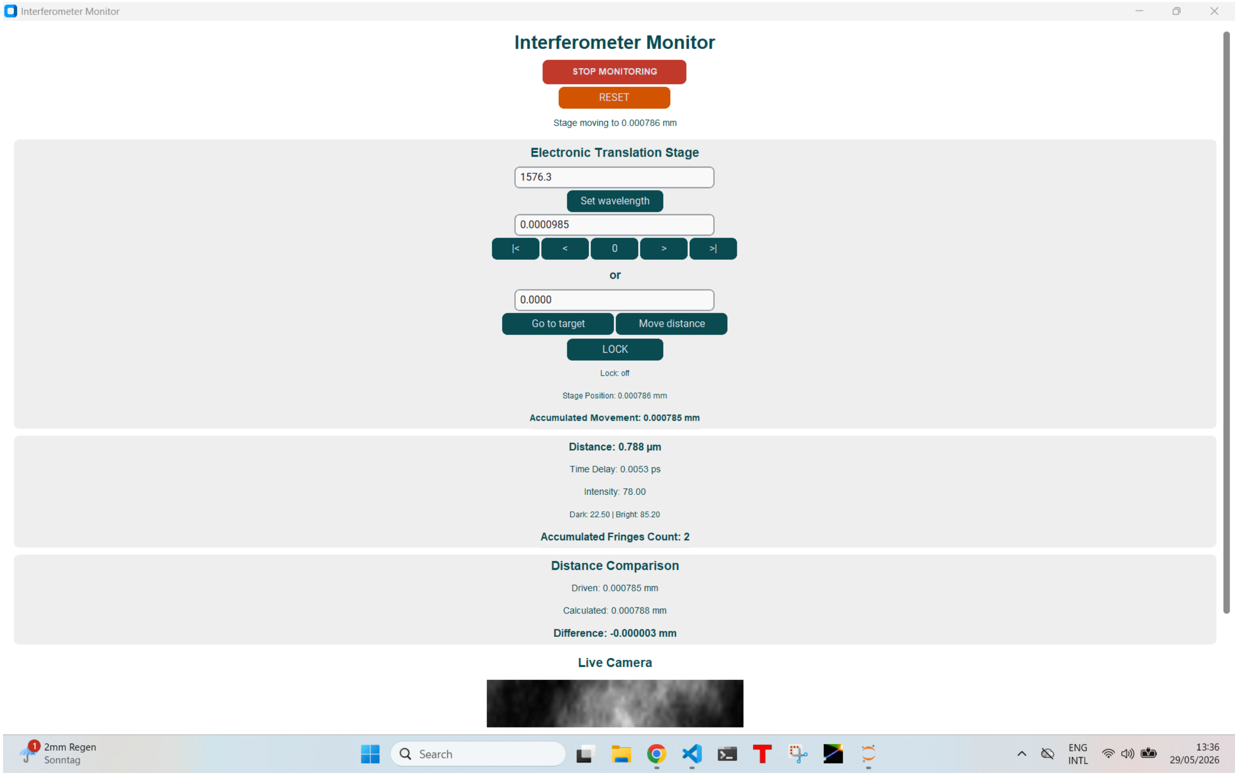

Figure 4.1: Graphical user interface of the developed Camera monitoring software.

Separating camera functionality from the main application simplified maintenance and reduced the complexity of the graphical user interface code.

4.1.3 Stage Handler

Control of the electronic translation stage was implemented in a separate `StageHandler` module. This module handles all communication with the stage hardware, including position queries, absolute movements, relative movements and motion limits.

4.1.4 Automatic Calibration

Reliable fringe detection requires accurate identification of bright and dark interference states. Since the observed intensity range depends on camera exposure, optical alignment, and environmental conditions, fixed threshold values are generally unsuitable.

To overcome this limitation, an automatic calibration procedure was implemented. Whenever monitoring is started, the software enters a calibration phase lasting five seconds.

During this period, the translation stage performs a predefined forward and backward motion of about one fringe distance while the software continuously records the measured fringe intensities. The resulting intensity values are stored and used to determine the minimum intensity I_{min} and the maximum intensity I_{max}.

The dark threshold is then calculated as

$$I_{\text{dark}} = I_{\text{min}} + 0.125\left(I_{\text{max}} - I_{\text{min}}\right), \tag{4.1}$$

while the bright threshold is determined by

$$I_{\text{bright}} = I_{\max} - 0.40\,(I_{\max} - I_{\min})\,. \tag{4.2}$$

These parameters were determined experimentally and were found to provide reliable fringe detection under the investigated operating conditions. The dark threshold was set close to $I_{\min}$, resulting in a conservative classification that accepts only clearly defined intensity minima. The bright threshold was set farther below $I_{\max}$, because the measured bright fringes did not always reach the absolute maximum intensity.

After calibration, the stage automatically returns to its reference position and all measurement counters are reset. To prevent accidental user interaction during this transition, all control buttons remain disabled for an additional two seconds.

As a fallback option, manually defined threshold values can also be used if automatic calibration fails.

4.1.4.1 Fringe Detection Algorithm

The core functionality of the software is the detection and counting of interference fringes from the camera signal.

4.1.4.2 Threshold-Based Detection

The measured fringe intensity is continuously compared with the calibrated dark and bright thresholds in a fixed region of interest in the camera signal. Whenever the intensity falls below the dark threshold, the software identifies the current fringe state as dark. Similarly, intensities above the bright threshold are classified as bright.

Using two separate thresholds reduces sensitivity to small fluctuations near the transition region.

4.1.4.3 Moving Average Filtering

Camera noise and short-term intensity fluctuations may cause individual frames to be incorrectly classified. To suppress such disturbances, the software applies a moving average filter to the most recent five intensity measurements.

The filtered intensity is given by

$$I_{\text{filtered}} = \frac{1}{N}\sum_{i=1}^{N} I_i, \tag{4.3}$$

where N is the number of averaged measurements and I_i represents the intensity of the i-th measurement. The filter window size is set to

$$N = 5. \tag{4.4}$$

This smoothing operation significantly improves detection stability without noticeably affecting response time.

4.1.4.4 State Machine

Fringe counting is implemented using a state-based approach. The software first waits until a stable dark fringe has been detected. Once this condition is satisfied, the system enters a waiting state and monitors for the appearance of a stable bright fringe.

A fringe is counted only when a confirmed bright state follows a previously confirmed dark state. After a fringe has been counted, the state machine is reset and the process begins again.

This approach prevents multiple counts from being generated by small oscillations around the threshold values.

4.1.4.5 Consecutive Frame Validation

To further improve robustness, a state transition is accepted only after multiple consecutive frames satisfy the corresponding threshold condition.

The values

$$N_{\text{dark}} = 3 \quad \text{and} \quad N_{\text{bright}} = 3 \tag{4.5}$$

were found experimentally to provide reliable operation.

Consequently, at least three consecutive dark frames are required before a dark state is accepted, and at least three consecutive bright frames are required before a bright state is accepted.

4.1.4.6 Cooldown Mechanism

The cooldown time was initially determined experimentally. For the stage velocity used during the experiment, a value of

$$t_{\text{cooldown}} = 80\,\text{ms} \tag{4.6}$$

was found to reliably prevent multiple counts caused by intensity fluctuations around the detection threshold.

However, a fixed cooldown time is only suitable for a limited range of stage velocities. If the user selects a higher velocity, the time between two consecutive fringes decreases. In this case, a fixed cooldown may suppress valid fringes.

Therefore, the cooldown time is calculated adaptively from the selected stage velocity. Instead of using a fixed delay, the software estimates the expected time interval between two consecutive fringes from the current velocity setting and the known laser wavelength. The experimentally determined cooldown of approximately 80 ms corresponded to about one eighth of the expected fringe period at the measurement velocity used during calibration. This ratio was therefore used as the scaling factor for the adaptive implementation:

$$t_{\text{cooldown}} = \frac{T_{\text{Fringe}}}{8}. \tag{4.7}$$

Here, T_{Fringe} denotes the velocity-dependent time between two consecutive fringes.

Thus, the cooldown time automatically decreases when the user selects a higher stage velocity and increases when a lower velocity is selected. This prevents valid fringes from being suppressed while preserving the experimentally determined relation between the cooldown time and the expected fringe period.

4.1.5 Translation Stage Control

The software provides several methods for controlling the electronic translation stage.

The user may move the stage to an absolute position, move it by a relative distance, or perform incremental movements using a configurable step size. Additional buttons allow direct movement to the minimum position, center position, or maximum position of the stage travel range.

The default step size is automatically calculated from the laser wavelength and corresponds to one quarter of the fringe distance. This allows precise positioning while ensuring that individual fringes cannot easily be skipped during motion.

All stage movements are executed in background threads to prevent the graphical user interface from becoming unresponsive during motion.

4.1.6 Conversion of Fringe Count to Displacement

The fringe count obtained from the interference signal is converted into a physical displacement using the known laser wavelength.

For a Michelson interferometer, the optical path difference changes by twice the mechanical displacement of the translation stage. Since the implemented algorithm counts transitions between dark and bright fringes, one counted fringe step corresponds to a stage displacement of

$$d_{\text{fringe}} = \frac{\lambda}{2}. \tag{4.8}$$

Using the measured wavelength of the selected laser source,

$$\lambda = 787.3\,\text{nm}, \tag{4.9}$$

the corresponding displacement per counted fringe step is

$$d_{\text{fringe}} = 0.394\,\mu\text{m}. \tag{4.10}$$

The total interferometrically measured displacement is calculated from the accumulated fringe count according to

$$d = N_{\text{fringe}} \cdot d_{\text{fringe}}, \tag{4.11}$$

where N_{fringe} denotes the accumulated fringe count. In addition, the software converts the measured displacement into the corresponding optical time delay:

$$\Delta t = \frac{2d}{c}, \tag{4.12}$$

where c is the speed of light.

4.1.7 Comparison of Stage and Interferometric Displacement

A key objective of the developed software is to compare the displacement reported by the translation stage controller with the displacement measured interferometrically by fringe counting.

For each movement, the software continuously calculates the displacement deviation

$$\Delta d = d_{\text{stage}} - d_{\text{interferometer}}, \tag{4.13}$$

where d_{stage} is the displacement reported by the stage controller and $d_{\text{interferometer}}$ is the displacement determined from the fringe count.

4.2 Software for Fringe Detection with the Photodiode

The software developed for the photodiode-based fringe detection follows the same general structure as the camera-based implementation described in the previous section. The graphical user interface is again implemented using the `CustomTkinter` library, while the hardware communication is separated into dedicated handler classes. Instead of processing camera images, the software now uses a `SingleDiodeHandler` class, which establishes the connection to the National Instruments data acquisition (NI-DAQ) device and continuously reads the voltage signal of the photodiode, and the main application, which coordinates the user interface, the measurement process and the control of the electronic translation stage.

As with the camera software, the measurement process is executed in a separate thread. This allows the voltage acquisition, translation stage movement and graphical user interface to run simultaneously without freezing the application. During the measurement, the user interface continuously updates the measured voltage, the accumulated fringe count, the calculated optical path difference, the corresponding time delay and the current position of the translation stage.

Figure 4.2 shows the user interface of the single-photodiode software. Because not all controls and plots fit into one visible window, the interface is shown in two parts.

4.2.1 Measurement Procedure

After the user starts the measurement, the software first establishes communication with the NI-DAQ device and the translation stage. Before the actual fringe detection begins, the first 200 acquired voltage samples are recorded while the translation stage remains stationary. These samples are averaged to determine the electrical baseline of the photodiode signal,

$$V_{\text{baseline}} = \frac{1}{N}\sum_{i=1}^{N} V_i, \tag{4.14}$$

with $N = 200$. This baseline is then subtracted from every following measurement,

$$V_{\text{clean}} = V_{\text{raw}} - V_{\text{baseline}}, \tag{4.15}$$

which removes the constant voltage offset and improves the signal-to-noise ratio.

Another version of the software also includes the option to use a reference photodiode. In this configuration, a small fraction of the laser beam is split off directly behind the laser output and detected by a second photodiode. The reference signal can then be used for normalization in order to compensate for slow fluctuations of the laser intensity.

4.2.2 Automatic Calibration

Before the measurement starts, the software performs an automatic calibration equal to the one used in the Camera software.

The photodiode signal showed noticeable fluctuations in both amplitude and offset over time. Therefore, detecting fringes by identifying local minima and maxima, as done for the camera images, did not provide reliable results. Instead, an adaptive calibration procedure was implemented.

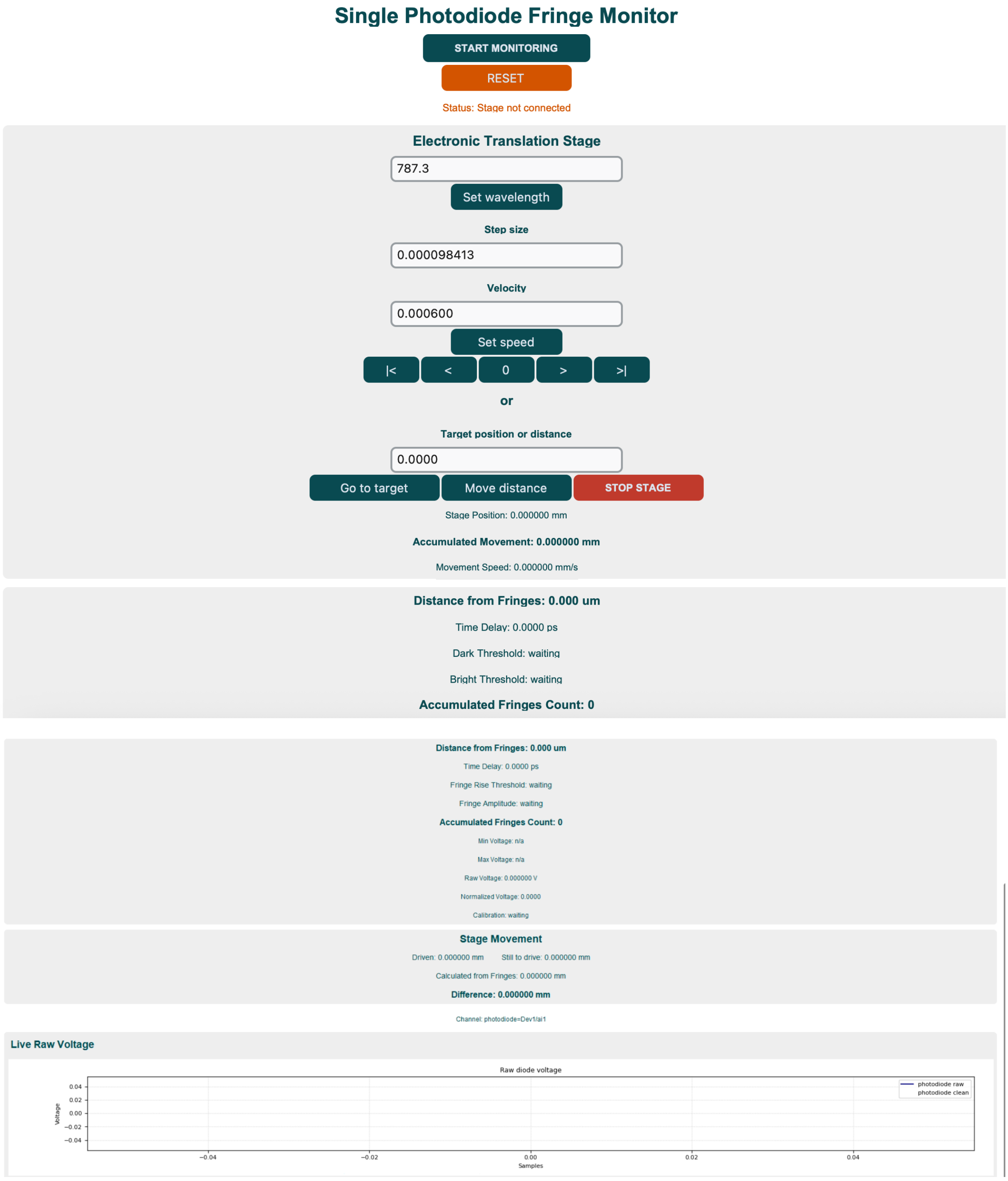

Figure 4.2: Graphical user interface of the single diode software. Top: upper part of the scrollable interface. Bottom: lower part of the interface.

First, the recorded calibration data are smoothed using a moving average filter to reduce high-frequency noise. Afterwards, local minima and maxima are identified from the filtered signal. From these extrema, the software estimates the fringe amplitude while automatically rejecting small extrema that originate from electrical noise. In addition, the noise level is estimated from the calibration data and used to calculate a minimum valid fringe amplitude. Only signal changes that clearly exceed this noise level are accepted as valid fringes.

Based on the measured fringe amplitude, the software automatically calculates the thresholds required for the subsequent fringe detection algorithm. Since these thresholds are determined individually for every measurement, the software adapts automatically to changing signal conditions without requiring manual parameter adjustment.

4.2.3 Fringe Detection

The implemented fringe detection algorithm is based on the measured fringe amplitude rather than on directly detecting local minima and maxima.

The detection algorithm then operates as a state machine consisting of a dark state and a bright state.

Initially, the algorithm waits until the signal reaches the dark region of an interference fringe. Once this condition is fulfilled, the signal is monitored until it rises by a predefined fraction of the calibrated fringe amplitude. When this threshold is exceeded for several consecutive samples, one fringe is counted.

The control of the translation stage as well as distance calculation, cooldown period and all further calculations are identical to the Camera Software mentioned above.

Finally, the software provides the possibility to record all measured data. The recorded dataset contains the relative measurement time, the raw voltage, the baseline-corrected voltage, the accumulated fringe count, the calculated optical distance and the current position of the translation stage. After the measurement, these data can be exported as a CSV file for further analysis. A visualization of the user interface is shown in Figure 4.2.

4.3 Software for Homodyne Quadrature Detection

The software for homodyne quadrature detection is based on the same structure as the software presented for the single-photodiode measurements. Therefore, components such as the graphical user interface, data acquisition via the NI-DAQ device, baseline correction, data recording and the general software architecture are implemented in the same way and are not described again in detail. The main difference is that the interferometric signal is now evaluated from two photodiodes instead of only one. These two signals are phase shifted by 90°, allowing the optical phase to be determined directly.

The analog voltages of both photodiodes are recorded simultaneously by the `NIPhotodiodeReader` class using two analog input channels of the NI-DAQ device. After acquisition, both signals are processed by the `HomodyneQuadratureCounter`. In parallel, a `SingleSignalFringeCounter` is executed for each signal. Its implementation is identical to the one described in the previous section and is mainly used to verify whether both photodiode signals contain visible interference fringes before the quadrature evaluation is started.

The graphical user interface of the homodyne quadrature software is shown in Figure 4.3. Since the interface contains more information than can be displayed in one window view, the upper and lower parts of the scrollable interface are shown separately.

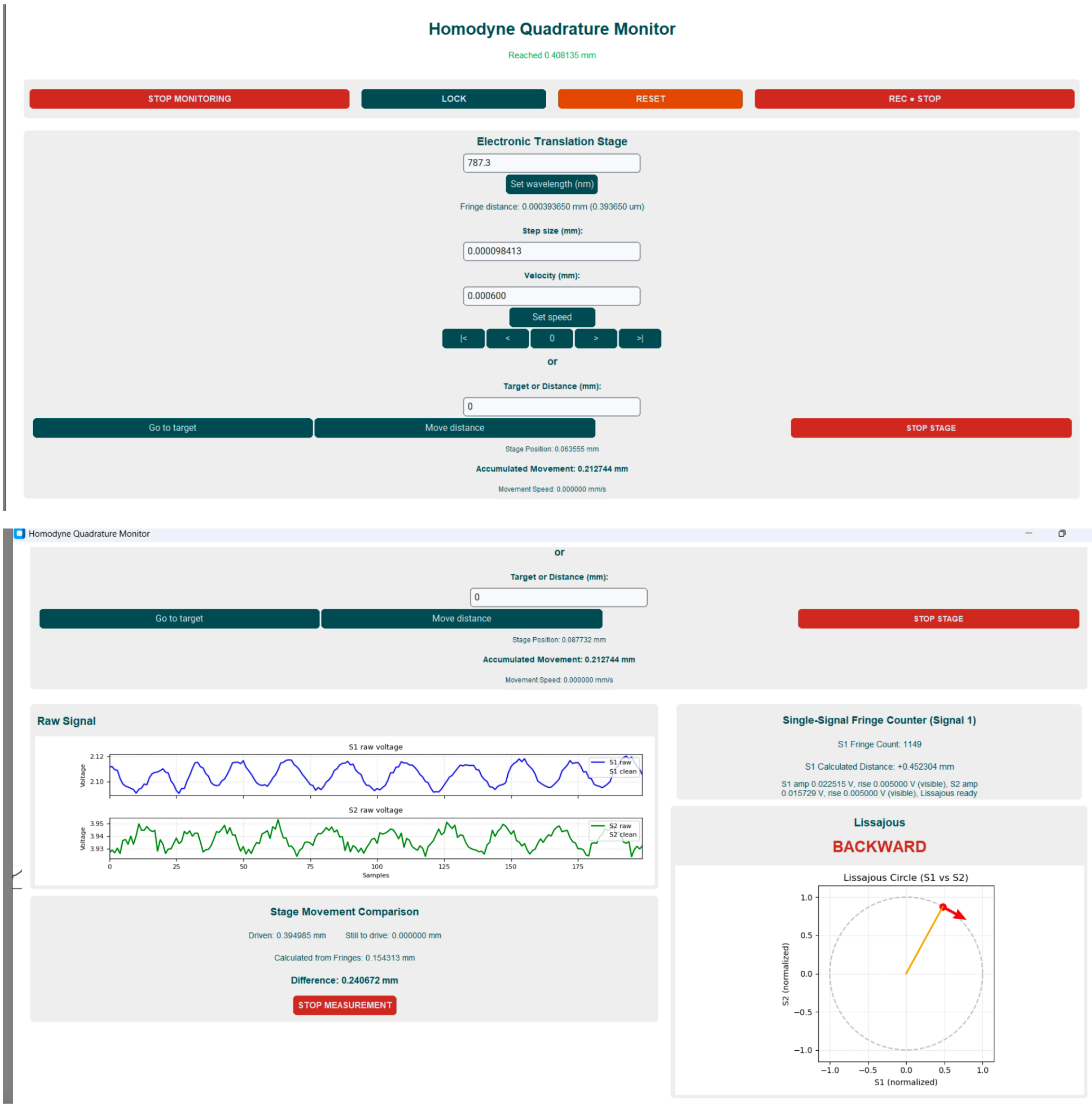

Figure 4.3: Graphical user interface of the homodyne quadrature software. Top: upper part of the scrollable interface. Bottom: lower part of the interface.

4.3.1 Calibration

Before each measurement, both photodiode signals are calibrated. During this process, the translation stage is moved over a short distance while several interference fringes are recorded. From these measurements, the minimum and maximum voltages of each signal are determined. The voltage offset is calculated as

$$V_{\text{offset}} = \frac{V_{\max} + V_{\min}}{2}, \tag{4.16}$$

while the normalization factor is given by

$$V_{\text{scale}} = \frac{V_{\max} - V_{\min}}{2}. \tag{4.17}$$

The normalized signals are then obtained by

$$S_{1,\text{norm}} = \frac{S_1 - V_{\text{offset},1}}{V_{\text{scale},1}}, \tag{4.18}$$

$$S_{2,\text{norm}} = \frac{S_2 - V_{\text{offset},2}}{V_{\text{scale},2}}. \tag{4.19}$$

This normalization compensates for different signal amplitudes and voltage offsets of both photodiodes and approximately maps the measured data onto a unit circle in the Lissajous representation.

The fringe count is then detected using only one of the photodiodes and the exact same fringe counter as in the Single Photo diode software.

4.3.2 Phase Determination

The phase difference between the two normalized signals is calculated using the four-quadrant arctangent,

$$\varphi = \text{atan2}\left(S_{2,\text{norm}}, S_{1,\text{norm}}\right). \tag{4.20}$$

[17]. Since the arctangent only returns phase values between $-\pi$ and π, discontinuities occur whenever the phase crosses this interval. To obtain a continuous phase signal, phase unwrapping is necessary. The phase difference between two consecutive measurements is first wrapped back into the interval $[-\pi, \pi]$ and then accumulated,

$$\varphi_{\text{unwrap}} = \varphi_{\text{unwrap}} + \Delta\varphi. \tag{4.21}$$

The accumulated phase is converted into the corresponding fringe position by

$$N_{\text{fringe}} = \frac{\varphi_{\text{unwrap}}}{2\pi}. \tag{4.22}$$

Finally, the optical displacement is calculated using the fringe distance,

$$d = N_{\text{fringe}} \cdot \frac{\lambda}{2}, \tag{4.23}$$

where λ is the laser wavelength.

4.3.3 Direction Detection

An advantage of the quadrature method is that the direction of movement can be determined directly from the sign of the phase change described in the previous section. To reduce the influence of noise, the software evaluates the average phase change over several samples instead of using a single value. If this average value is larger than the switching threshold of 0.003 rad, the motion is classified as forward. If it is smaller than −0.003 rad, the motion is classified as backward. For smaller absolute phase changes, the stage is treated as stationary.

A lower release threshold of 0.001 rad prevents short noise spikes from causing rapid switching between forward, backward, and stationary states. These values were also determined experimentally.

4.3.4 Lissajous Representation

To visualize the measurement, the normalized photodiode signals are displayed as a Lissajous figure. Ideally, both signals form a circular trajectory, where the angular position corresponds directly to the optical phase [18]. During operation, the current measurement point, the recent trajectory and the detected direction of rotation are updated continuously.

Since the raw photodiode signals are affected by electrical noise, the displayed circle is additionally smoothed. For this purpose, the phase of the last measurement points is approximated by a linear fit. The fitted phase is then transformed back into sine and cosine functions, resulting in a smooth circular trajectory.

4.3.5 Position Lock

One limitation of the translation stage used in this work was that it did not remain perfectly stationary after reaching the commanded position. Instead, small mechanical drifts around the target position were observed. To compensate for these drifts, an active position lock was implemented.

The position lock can only be activated when no measurement is running. Likewise, once the lock is active, measurement mode and manual stage movements are disabled to avoid conflicts between user commands and the feedback controller.

When the user activates the lock, the current optical phase and the current height of the fringe count is defined as the reference position. If a previous calibration has already been performed, for example during a measurement, the calibrated fringe amplitudes are reused. Otherwise, the software prompts the user to enter an expected fringe amplitude manually. This value is then used as the detection threshold.

During operation, the software continuously evaluates both the single-photodiode fringe counter and the homodyne phase measurement. A correction is only initiated if two conditions are fulfilled simultaneously. First, the detected fringe must exceed the calibrated fringe threshold to avoid false detections caused by noise. Second, the homodyne evaluation must provide a valid movement direction. Only when both criteria are satisfied is the detected displacement interpreted as a real drift of the interferometer.

The required correction distance is calculated from the number of detected fringes and the corresponding fringe spacing. The translation stage is then commanded to move by the same distance in the opposite direction, thereby compensating the measured drift and returning the optical path length towards the reference position.

In practice, the achievable locking performance is limited by the mechanical behaviour of the translation stage. During every correction movement, additional interference fringes are generated because the stage itself is moving. These fringes are detected by the software and may trigger further correction movements before the previous correction has completely settled. Consequently, successive corrections are initiated although the original disturbance has already been compensated. With the positioning accuracy and response time of the available translation stage, this behaviour could not be completely eliminated. Therefore, the implemented position lock demonstrates the general feedback concept, but its long-term stability remains limited by the mechanical characteristics of the stage.

4.3.6 Software Optimizations

Due to the increasing number of implemented features, the computational workload of the software also increased. As a result, the graphical user interface became progressively less responsive. Although this issue could not be eliminated completely, mainly because of the large amount of data that had to be processed and displayed simultaneously within a Tkinter-based interface, several optimizations were implemented to significantly improve the responsiveness of the software.

The most important optimization was the extensive use of multithreading. Time-consuming tasks were executed in separate background threads, allowing the graphical user interface to continue updating independently. In particular, communication with the NI-DAQ device and the translation stage involves blocking hardware operations, which would otherwise freeze the user interface while waiting for a response. Therefore, all measurement routines, stage movements and hardware communication were executed in dedicated background threads, while the main thread was reserved exclusively for updating the graphical interface and processing user input.

The photodiode signals are sampled every

$$\Delta t_{\text{sample}} = 5 \text{ ms}, \tag{4.24}$$

which corresponds to a sampling frequency of

$$f_{\text{sample}} = 200 \text{ Hz}. \tag{4.25}$$

This sampling rate was found to provide sufficient temporal resolution for fringe detection while keeping the computational load low.

Updating every graphical element at the sampling rate would unnecessarily increase the CPU usage and noticeably reduce the responsiveness of the interface. Therefore, the measurement thread only stores the newest acquired sample, while the graphical user interface is refreshed independently every

$$\Delta t_{\text{UI}} = 100 \text{ ms}, \tag{4.26}$$

corresponding to an update frequency of

$$f_{\text{UI}} = 10 \text{ Hz}. \tag{4.27}$$

For the live voltage plots, only the most recent 200 samples are displayed. At a sampling frequency of 200 Hz, this corresponds to a visible time window of approximately

$$T_{\text{plot}} = 200 \cdot 5 \text{ ms} = 1 \text{ s}. \tag{4.28}$$

The movement status of the translation stage is updated every

$$\Delta t_{\text{stage}} = 100 \text{ ms}, \tag{4.29}$$

which is sufficient because the mechanical motion of the stage is several orders of magnitude slower than the photodiode sampling. During continuous stage movements, the position and velocity are recalculated every 50 ms, providing smooth feedback without overloading the communication interface.

To further reduce computational effort, the live display is updated using a buffering mechanism. Newly acquired samples are stored temporarily and only the most recent sample is transferred to the graphical interface during the next scheduled update. Consequently, multiple measurement samples can be acquired between two graphical updates without increasing the frequency.

CHAPTER 5

DEVELOPMENT OF THE INTERFEROMETER

With the software implementation completed, all components required for the interferometric measurements were available. This chapter presents the complete experimental setup used in this work.

5.1 Experimental Setup

The interferometer was initially assembled on an optical table using a manual translation stage in order to validate the measurement principle. After successful verification of the setup, the manual stage was replaced by the Thorlabs LTS300C/M automated translation stage for subsequent investigations.

A key objective during the development process was the identification of a suitable method for detecting the interference signal. Two different approaches were therefore investigated and compared. In the first approach, the interference pattern was recorded using a camera and evaluated through fringe detection. In the second approach, the interference signal was converted into an electrical signal using a photodiode. The performance of both methods was assessed with respect to signal quality and suitability for automated displacement measurements.

5.2 Fringe Detection Using a Camera

The first detection method was based on direct observation of the interference fringes using a Thorlabs CS165MU/M camera. Since the optical power incident on the camera exceeded the recommended operating range of the sensor, three absorptive neutral density filters with an optical density of (ND = 1.0) each were placed in front of the camera.

The complete experimental setup, including the automated translation stage, is shown in Figure 5.1. After careful alignment of the interferometer arms, stable and high-contrast interference fringes were observed in the ThorCam software, as shown in Figure 5.2.

The stability and visibility of the observed interference pattern indicated that image-based fringe detection was a promising approach.

5.2.1 Comparison with the Best HeNe Laser Source

As discussed in Chapter 3, the Thorlabs diode laser exhibited the most stable intensity profile and was therefore mainly used for the interferometric measurements. To compare its performance with a HeNe source, the camera-based fringe detection measurement described above was repeated using the Uniphase 1023P HeNe laser. This laser was selected because it showed the best intensity stability among the investigated HeNe sources, which was considered the most important criterion for reliable fringe detection.

The experimental setup and evaluation procedure were kept unchanged; only the laser source was replaced. Compared with the diode laser, the fringe motion was less clearly

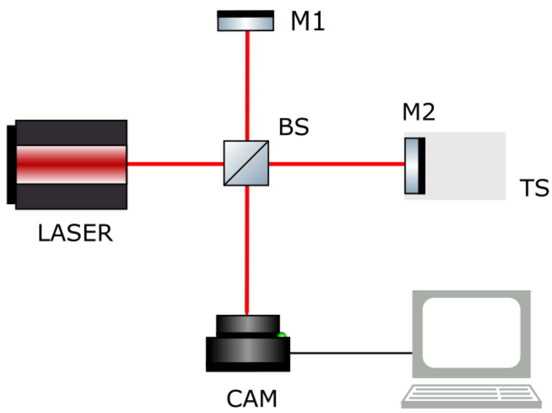

Figure 5.1: Schematic Representation of Michelson interferometer used for displacement measurements with camera detection. BS denotes the beam splitter, M1 the reference mirror, M2 the measurement mirror, TS the translation stage, and CAM the camera.

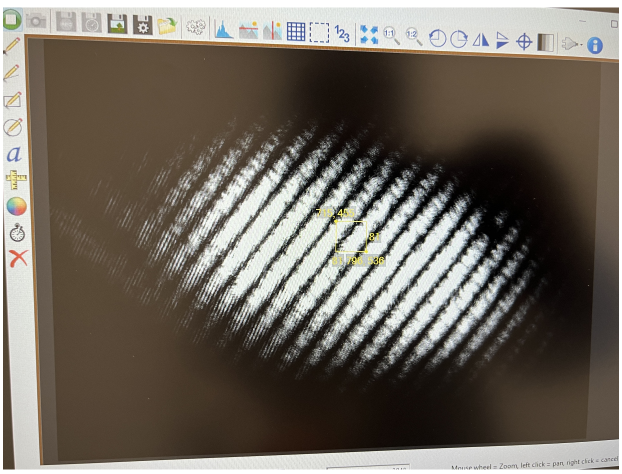

Figure 5.2: Stable interference fringes observed with the camera in the ThorCam software.

detectable when using the HeNe laser. In contrast, the diode laser produced fringes with sharper edges and higher temporal stability, which made the fringe motion easier to identify reliably. The corresponding camera-based evaluation result for the HeNe laser is shown in Figure 5.3.

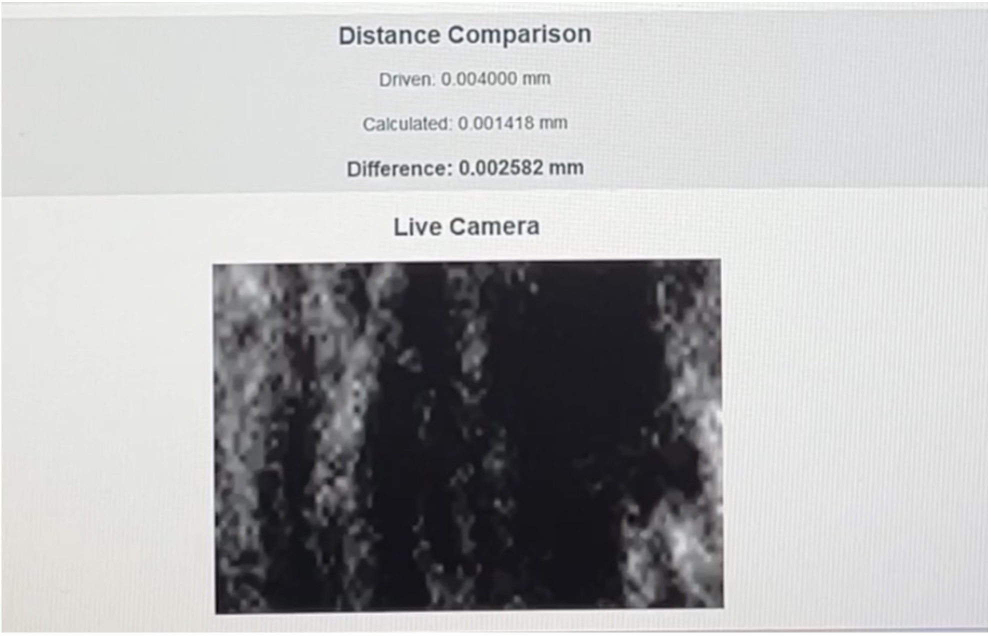

Figure 5.3: Camera-based fringe detection result obtained with the Uniphase 1023P HeNe laser.

5.3 Electronic Signal Detection

As an alternative to image-based fringe detection, an electronic detection approach was investigated. In this configuration, the optical output of the interferometer was directed onto a photodiode, converting the optical intensity variations into an electrical signal. The resulting signal was analyzed using a National Instruments USB-6221 data acquisition device. The experimental setup is shown in Figure 5.4.

In principle, displacement of the translation stage should produce periodic intensity variations in the interference signal, with destructive interference corresponding to intensity minima and constructive interference corresponding to intensity maxima.

The measurements were performed using the Thorlabs LTS300C/M automated translation stage. Since the interferometric detection principle remained consistent, the results can be directly compared with the camera-based diode laser measurements.

With these modifications, fringes could also be detected from the electronic photodiode signal. The corresponding result is shown in Figure 5.5.

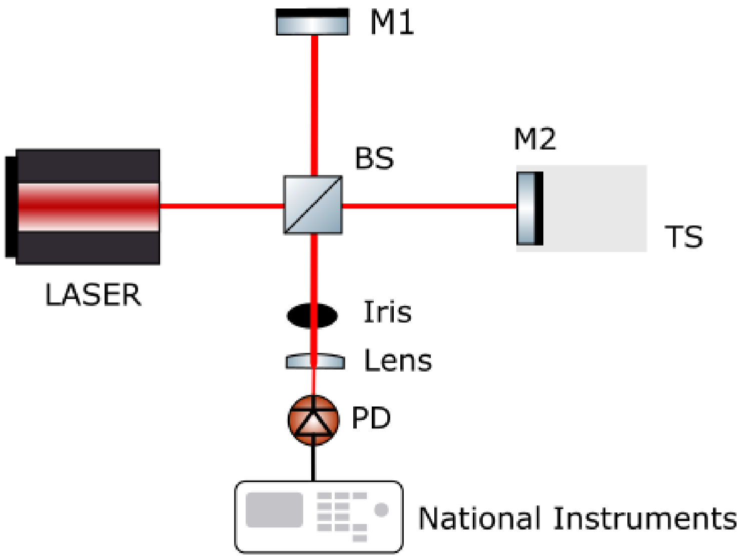

Figure 5.4: Electronic signal detection using a photodiode, schematic representation. BS denotes the beam splitter, M1 the reference mirror, M2 the measurement mirror, TS the translation stage, and PD the photodiode.

Figure 5.5: Fringe detection result obtained from the electronic photodiode signal.

5.4 Comparison of Detection Methods

The two investigated detection approaches showed different performance levels. The camera provided stable and clearly visible interference fringes with a high signal-to-noise ratio, enabling robust image-based evaluation. Furthermore, the recorded fringe patterns offered direct visual feedback during alignment and facilitated the development of automated analysis software.

The photodiode-based detection approach also produced a usable periodic signal. One advantage of this method is that the sinusoidal intensity variation can be monitored directly in the voltage signal. This makes missed fringes or undetected peaks easier to identify than in camera-based detection, where fringes may become difficult to distinguish during rapid motion. However, the photodiode signal was not consistently stable. In some intervals, the sinusoidal modulation flattened or shifted, and the number of visible fringes varied over comparable time periods. These observations indicate that the signal-to-noise ratio and baseline stability were not sufficient for the highest measurement accuracy.

Unfortunately, the absolute measurement accuracy of the camera- and photodiode-based methods cannot be compared quantitatively. Such an analysis would require a reference system with a significantly higher accuracy, for example a calibrated interferometer or a translation stage whose displacement is known with negligible uncertainty. Neither of these was available in this work.

Using the translation stage as the reference would not be appropriate, since the purpose of the developed interferometer is precisely to characterize and improve the stage's positioning accuracy. Any uncertainty in the stage motion would therefore directly affect the calculated measurement uncertainty, resulting in a circular validation.

Therefore, the two methods were compared in terms of their repeatability rather than their absolute accuracy. The translation stage was commanded to perform the same nominal displacement of 0.5 mm repeatedly, first using the camera-based system and then the photodiode-based system. The resulting fringe counts and calculated displacements were evaluated by comparing their standard deviations. Although this approach does not provide the absolute measurement error of either method, it allows their measurement repeatability to be assessed under identical experimental conditions and represents the most meaningful comparison that could be performed with the available experimental setup.

The repeatability comparison is shown in Figure 5.6. Across the three repeated measurements, the camera-based evaluation showed a slightly smaller variation in the calculated displacement than the photodiode-based evaluation. However, because only three measurements were performed, this difference cannot be considered statistically significant. Instead, the results indicate that both methods were able to reproduce the commanded displacement reliably under the investigated conditions.

A more noticeable difference can be seen in Figure 5.7. The camera-based signal was less affected by environmental noise and short-term disturbances, while the photodiode signal showed larger fluctuations and variations in the baseline. Although the repeatability measurements did not show a significant difference between the two methods, the camera-based approach provided a more stable signal under the investigated conditions.

During the experimental work, additional limitations of the experimental setup became apparent. The signal quality showed noticeable variations between repeated measurements. In addition to differences between individual measurements, the signal quality also varied from one day to another.

Another observation was that the interference signal occasionally required a short set-

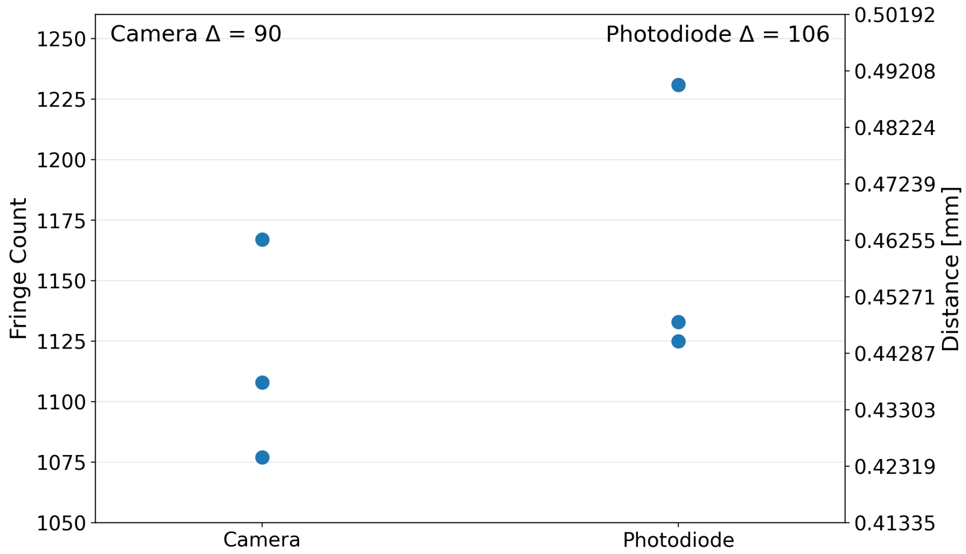

Figure 5.6: Comparison of the calculated displacements obtained from camera-based and photodiode-based fringe detection for repeated nominal displacements of 0.5 mm.

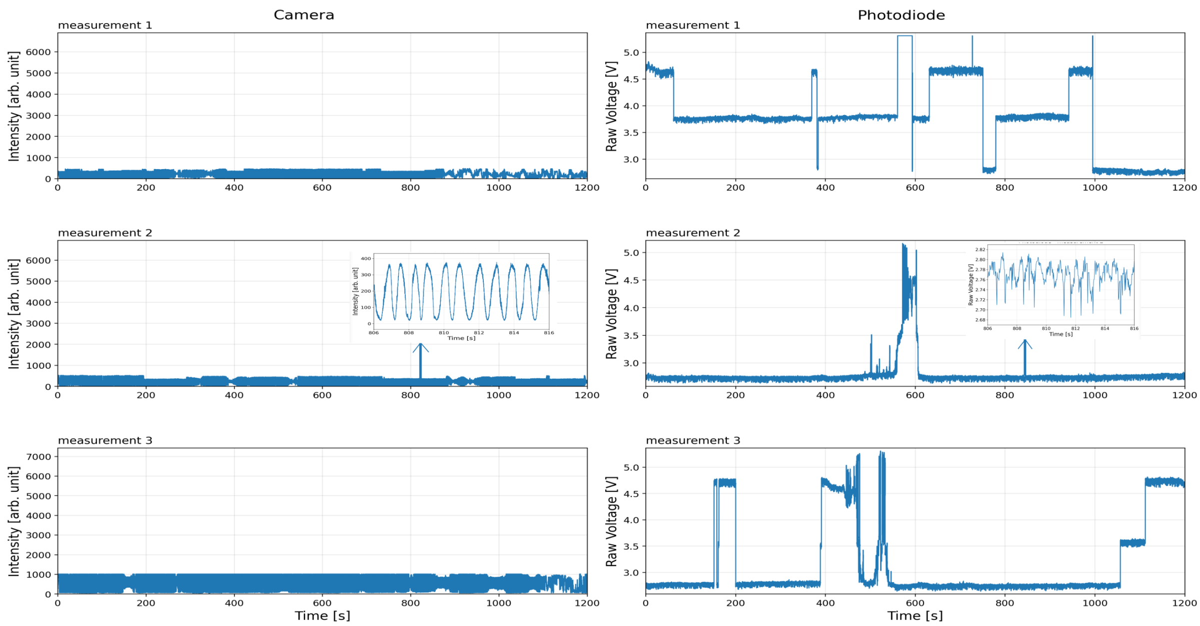

Figure 5.7: Side-by-side comparison of camera-based and photodiode-based fringe detection. The camera-based evaluation is less affected by environmental noise, while the photodiode signal exhibits stronger fluctuations. The camera-based measurement was performed at a stage velocity of 0.0004 mm s^{-1}, whereas the photodiode-based measurement was performed at 0.0006 mm s^{-1}. For clearer visualization, both datasets are shown only up to 1200 s, although the camera-based measurement contained additional data. This time interval was sufficient for comparing the signal stability of both detection methods.

tling time before stable fringes appeared. In these cases, the translation stage had already started moving, while the interference fringes became visible only after a short delay. The reason for this behaviour could not be identified within the scope of this thesis.

Overall, the system did not provide the same signal quality under all measurement conditions. If the setup is used in future experiments, it should be operated in a better isolated environment to reduce the influence of air currents, vibrations of the optical table, and other external disturbances. Due to the limited scope and timeframe of this thesis, implementing these improvements was not possible.

5.5 Quadrature Homodyne Interferometer

The interferometer was not only intended to improve the accuracy of the translation stage displacement measurement. A second important requirement was the detection of the direction of motion, since small forward and backward movements of the stage in a stationary position must be compensated for in order to enable active correction. For this reason, the Michelson interferometer was extended with polarization optics to form a quadrature homodyne interferometer. The corresponding laboratory setup and schematic representation are shown in Figure 5.8.

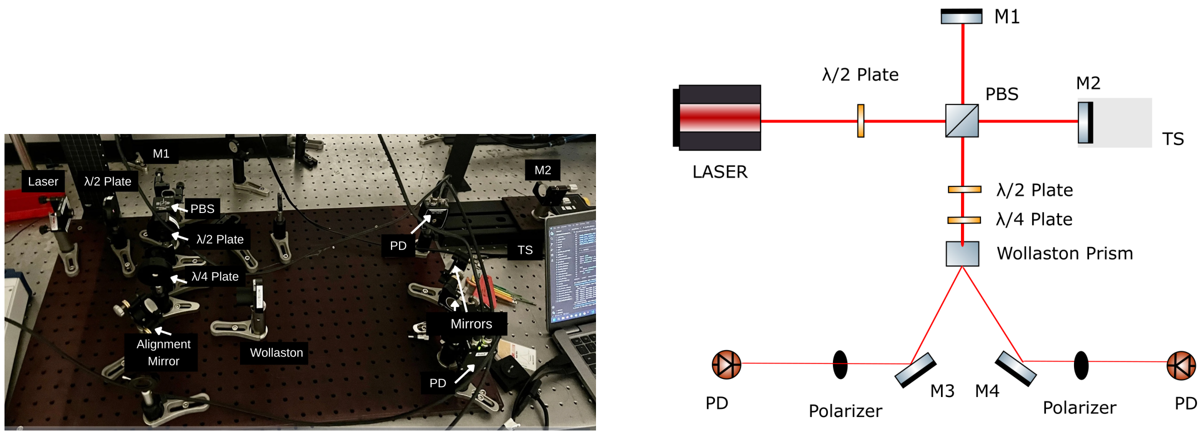

Figure 5.8: Laboratory setup and schematic representation of the quadrature homodyne interferometer. The setup extends the Michelson interferometer with polarization optics in order to generate two phase-shifted photodiode signals for direction-sensitive displacement detection. M1–M4 denote mirrors, PD denotes the photodiode, TS denotes the translation stage, and PBS denotes the polarizing beam splitter. Although the polarizers at the end are included in the schematic for completeness, they are omitted in the actual setup photograph, as the Wollaston prism theoretically acts as a perfect polarizing beam splitter.

In this configuration, the laser beam first passes through a half-wave plate, which rotates the linear polarization to 45°. The beam then enters a polarizing beam splitter, where it is ideally divided into equal s- and p-polarized components. These two components form the two interferometer arms and are reflected by mirrors.

After propagating through the interferometer arms, the two beams are recombined and pass through a second half-wave plate. This wave plate rotates the polarization by 45°, so that both beams contain equal amounts of s- and p-polarized light. The beams then pass through a quarter-wave plate, which introduces a phase shift of $\pi/2$ between the two polarization components and converts the light into circularly polarized light. This allows the Wollaston prism to generate two interference signals that are 90° out of phase, enabling quadrature detection. The Thorlabs MgF_2 (10 mm thickness) Wollaston prism acts as a polarization beam splitter and generates two output beams with orthogonal polarization components. These beams are redirected by two mirrors and then pass through polarizers to clean up the respective polarization states before detection with a photodiode.

The polarization states in the quadrature homodyne interferometer are illustrated schematically in Figure 5.9. This representation clarifies how the polarization optics generate two detection signals with an approximate phase shift of 90°.

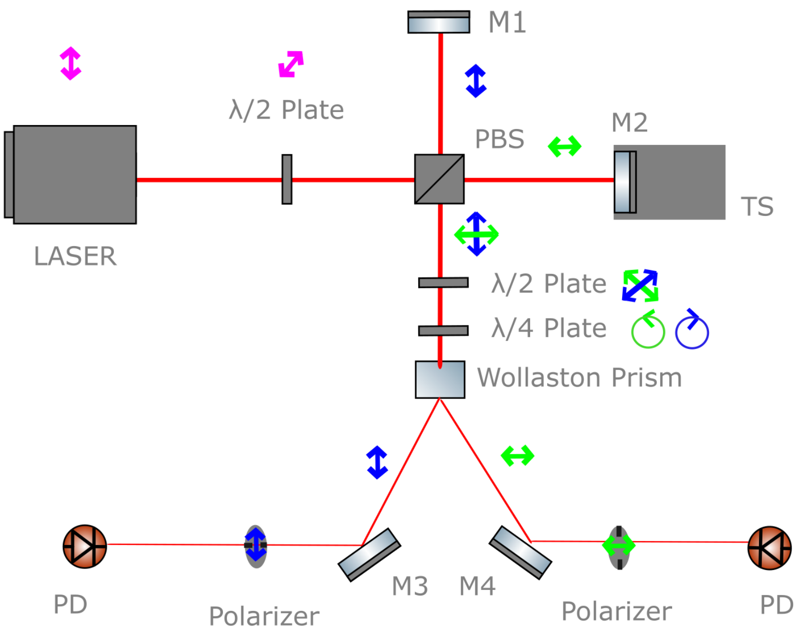

Figure 5.9: Schematic representation of the polarization states in the quadrature homodyne interferometer. M1–M4 denote mirrors, PD denotes the photodiode, TS the translation stage, and PBS the polarizing beam splitter. The rotation direction introduced by the $\lambda/4$ plate was chosen arbitrarily.

The two phase-shifted photodiode signals were successfully detected. The direction of motion can be determined from the Lissajous plot shown on the right-hand side of the graphical user interface (see Figure 4.3 in Chapter 4). The two quadrature signals form a circular trajectory, and the direction in which this trajectory is moving indicates the direction of motion.

A detailed evaluation of the recorded quadrature signals is shown in Figure 5.10. As indicated by the vertical reference line passing through corresponding points of the two fitted signals, the curves exhibit a phase shift of 90°. This confirms that the optical setup generated the required quadrature phase relation for direction-sensitive displacement detection.

To verify that the quadrature homodyne interferometer can determine the direction of motion over longer travel distances, the translation stage was moved repeatedly over a distance of 0.5 mm. This distance was chosen to be consistent with the comparison of the camera- and photodiode-based detection methods. At a stage velocity of $0.0006\,\mathrm{mm\,s^{-1}}$, this displacement corresponds to a comparatively long measurement time and therefore provides a useful test of the stability of the direction detection.

The stage was moved three times in the forward direction and three times in the back-

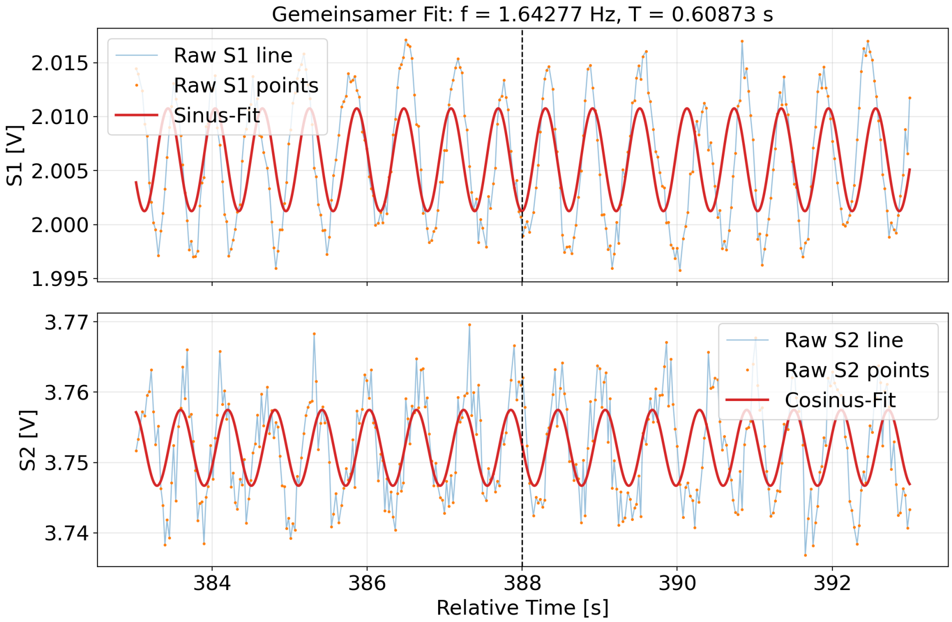

Figure 5.10: Detailed evaluation of the quadrature homodyne signals. The graph visualizes the sine and cosine photodiode signals used to determine the direction of motion. A minor linear trend, likely originating from the photodiode, was subtracted. The fitted curves show a phase shift of 90°, confirming the quadrature relation between the two signals.

ward direction. During each measurement, the software assigned every recorded data point to one of the three states *forward*, *backward*, or *still*. The resulting percentages are summarized in Table 5.1.

Table 5.1: Direction classification obtained with the quadrature homodyne interferometer during repeated 0.5 mm stage movements.

Commanded motion	Forward (%)	Backward (%)	Still (%)
Backward 1	18.93	63.82	17.25
Backward 2	17.08	66.27	16.65
Backward 3	12.68	70.79	16.54
Forward 1	67.01	13.08	19.91
Forward 2	63.94	17.82	18.24
Forward 3	78.55	8.22	13.23

For all six measurements, the largest fraction of data points corresponds to the commanded direction of motion. This shows that the quadrature homodyne interferometer was able to detect the direction correctly. However, a noticeable fraction of the data points was classified as *still* or as the opposite direction. This is mainly attributed to noise and signal instabilities. Therefore, the measurement confirms the general functionality of the direction detection, while also showing that further optimization of the signal quality would improve its robustness.

CHAPTER 6
INTEGRATION INTO THE FINAL SETUP

The final step of this thesis was the integration of the developed interferometric measurement system into the larger electro-optic sampling setup. The aim was to apply the interferometer to the translation stage used in the existing setup, since this stage was the component whose displacement accuracy had to be evaluated and improved.

Direct access to the translation stage inside the electro-optic sampling setup was limited because it was surrounded by several optical components. Therefore, the interferometer could not simply be placed directly around the stage. Instead, the optical signal reflected from the mirror mounted on the translation stage was guided across the optical table by several additional mirrors and then coupled into the developed interferometer setup.

To make this possible without permanently changing the original interferometer configuration, a flip mirror was added to the beam path. When the flip mirror is raised, the beam from the electro-optic sampling setup is directed into the interferometer and the displacement of the integrated translation stage can be measured. When the flip mirror is lowered, the interferometer can still be used in its original configuration with the separate test translation stage. This allowed switching between the standalone interferometer setup and the integrated measurement configuration.

Figure 6.1 shows both configurations.

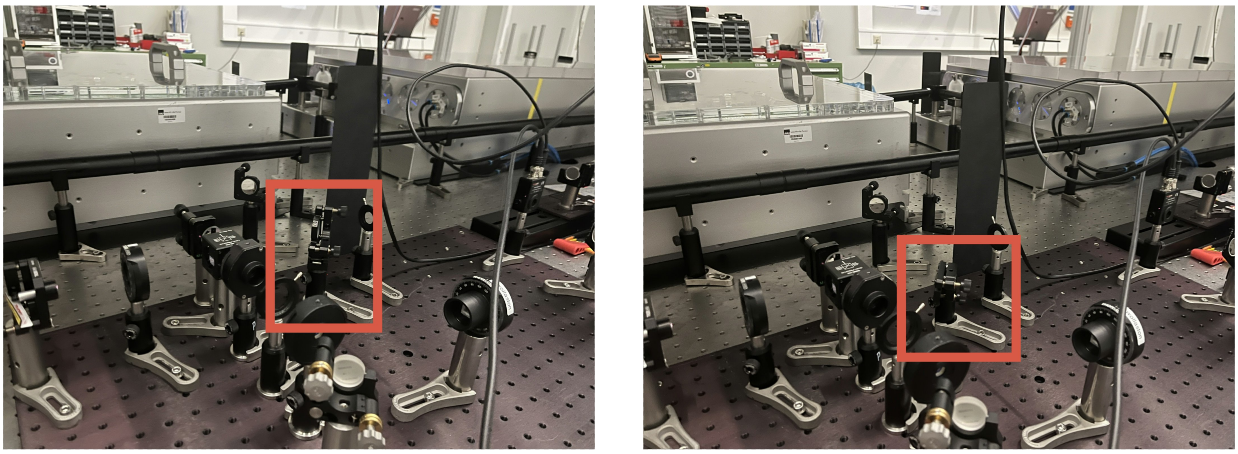

Figure 6.1: Integration of the interferometer into the electro-optic sampling setup. Left: integrated configuration with the flip mirror raised, so that the signal from the external setup is coupled into the interferometer. Right: standalone configuration with the flip mirror lowered, so that the interferometer can be used with the local translation stage.

Due to the limited timeframe of this thesis, the integrated configuration could not be experimentally tested. However, since the interferometric measurement principle and the optical detection path remain unchanged, the setup is expected to be suitable for displacement measurements after careful alignment and calibration.

CHAPTER 7 CONCLUSION

7.1 Final System

Based on the results presented in this thesis, the combination of the Thorlabs diode laser and the camera-based detection system proved to be the most stable choice, as it was significantly less affected by environmental noise. This stability is likely because the camera averages the optical signal over its exposure time, whereas the photodiode captures more high-frequency noise due to its higher sampling rate. However, since a key objective of the developed interferometer was to determine the direction of motion, the quadrature homodyne setup should be integrated into the final system. Although some signal fluctuations remain visible, it reliably provides a clear and direction-sensitive measurement signal.

The direction-sensitive measurement itself was successfully demonstrated. The resulting active position lock (Section 4.3.5), however, did not reach stable long-term operation, since successive correction movements of the translation stage repeatedly generated new interference fringes before the previous correction had settled. While the direction of motion could therefore be reliably determined and corrected for, achieving a fully stable closed-loop position lock remains an open task.

7.2 Outlook

The interferometric displacement measurement system developed in this thesis provides a solid basis for future work. Although the system was successfully implemented, several aspects could be further improved.

One important step would be to increase the stability of the experimental setup. During the measurements, environmental influences such as vibrations affected the interference signal. A better isolated setup would reduce these disturbances and improve the measurement reliability.

In addition, the origin of the day-to-day variations observed for the diode laser should be investigated. Identifying and eliminating the cause of these fluctuations could further improve the stability of the system.

Furthermore, the stability of the active position lock should be improved, for example by better decoupling the correction movements of the translation stage from the fringe detection, or by using a translation stage with faster and more precise settling behavior.

Finally, the complete measurement system should be tested in the intended experimental setup.

APPENDIX A

USE OF ARTIFICIAL INTELLIGENCE TOOLS

For transparency, this section describes how artificial intelligence tools were used during the preparation of this thesis.

Claude Opus 4.8, Gemini 3.1 Pro and ChatGPT 4o mini were used to improve the fringe-counting software. They helped with finding and fixing programming errors, making the code more robust, and adding safety checks in order to prevent the expensive hardware from damage. The main code structure and the underlying logic were developed by the author.

These tools were also used to improve the wording and readability of parts of the written text, proofreading and language suggestions. They were not used to independently create scientific content or citations.

APPENDIX B

OPTICAL COMPONENTS

For the sake of completeness, this appendix lists the optical and electrical components used throughout this work. For some components, the exact model numbers could not be verified because they were already pre-mounted in their respective holders.

Table B.1: Optical and electrical components used in the experimental setups.

Component	Manufacturer	Model Number
Spectrometer (300–1100 nm)	Ocean Optics	SR6VN500-10
Optical fiber	Thorlabs	LA4130UV
Silver mirrors	Thorlabs	–
Diode laser	Thorlabs	CPS780S
Helium-Neon laser	Uniphase	1023P
Helium-Neon laser	Uniphase	1103P (SN: 1177761)
Helium-Neon laser	Uniphase	1103P (SN: 1108380)
Helium-Neon laser	Uniphase	1122P
Helium-Neon laser	Uniphase	1507P
Lens (CaF_2, $f = 75\,\mathrm{mm}$)	Thorlabs	–
Lens (CaF_2, $f = 50\,\mathrm{mm}$)	Thorlabs	–
Beam profiling camera	Cinogy	CinCam CMOS-120
Power meter console/detector	Ophir Vega	7Z01560
50:50 beam splitter (400–700 nm)	Thorlabs	CCM1-BS013/M
Camera	Thorlabs	CS165MU/M (SN: 37615)
Automated translation stage	Thorlabs	LTS300C/M
Data acquisition (DAQ) device	National Instruments	USB-6221
Half-wave plate ($\lambda/2$, 894 nm)	B. Halle	–
Half-wave plate ($\lambda/2$, 400–850 nm)	Thorlabs	–
Quarter-wave plate ($\lambda/4$, 600–2700 nm)	B. Halle	–
Polarizing beam splitter (420–650 nm)	Thorlabs	CCM1-PBS251/M
Photodiode (Si)	Thorlabs	PDA100A-EC
Photodiode (InGaAs)	Thorlabs	PDA10CS-EC
Wollaston prism (MgF_2, 10 mm thickness)	Thorlabs	WPM10
Manual flip mount	Thorlabs	–

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SELBSTSTÄNDIGKEITSERKLÄRUNG

Hiermit erkläre ich, die vorliegende Arbeit selbstständig verfasst und keine anderen als die in der Arbeit angegebenen Quellen und Hilfsmittel verwendet zu haben.

München, den 28.07.2026.

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Carolina Krotsch

DECLARATION OF AUTHORSHIP

Hereby, I declare that I have written this thesis by myself and that I have not used any sources or aids other than those indicated in this thesis.

Munich, the 28.07.2026.

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